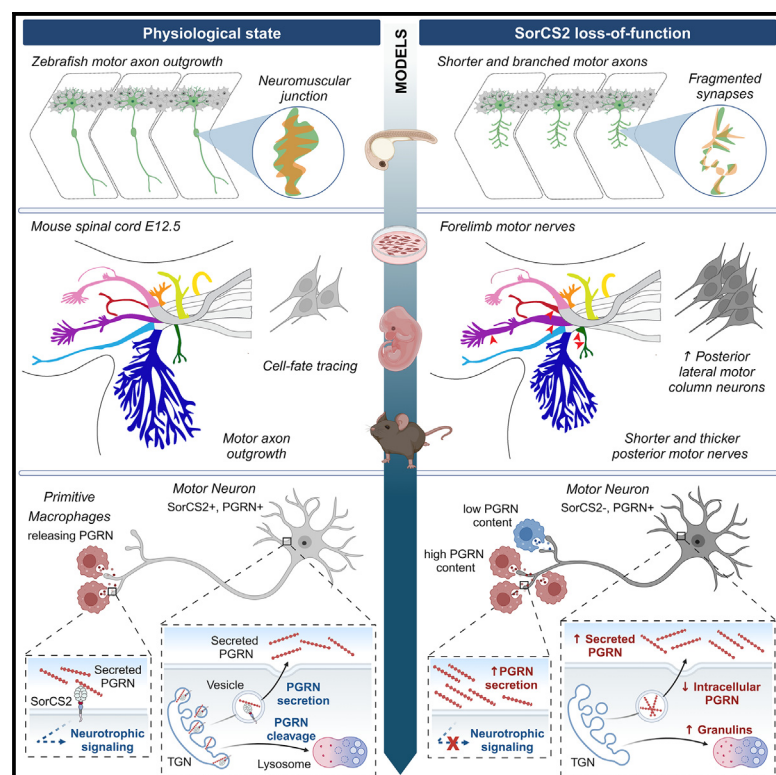


SorCS2 binds progranulin to regulate motor neuron development

Graphical abstract



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In brief

Thomassen et al. use various vertebrate models to assess the function of SorCS2 receptor in motor neuron development and regeneration. They describe that SorCS2 regulates motor neuron identity, synapse architecture, and motor nerve innervation via its binding to progranulin. Intracellularly, this interaction further determines progranulin secretion, sorting, and cleavage.

Highlights

- SorCS2 controls motor neuron cell-fate decisions, axon outgrowth, and synapse morphology
- SorCS2 deficiency impairs behavior of primitive macrophages associated with motor axons
- SorCS2 is a PGRN receptor required for its neurotrophic signaling in motor neurons
- SorCS2 binds PGRN to regulate its trafficking, secretion, and conversion to mature granulin



Article

SorCS2 binds progranulin to regulate motor neuron development

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SUMMARY

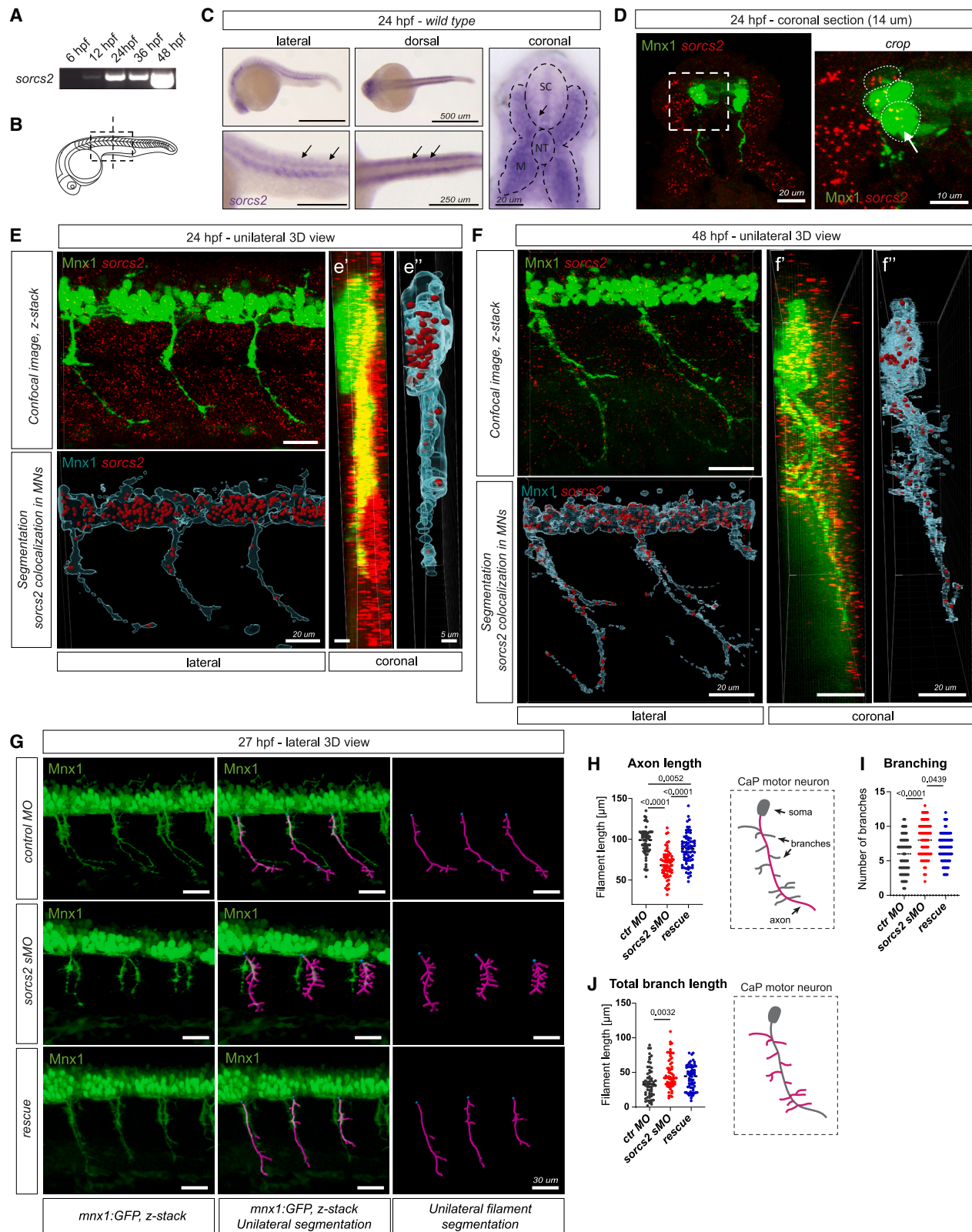
Motor neuron (MN) development and nerve regeneration requires orchestrated action of a vast number of molecules. Here, we identify SorCS2 as a progranulin (PGRN) receptor that is required for MN diversification and axon outgrowth in zebrafish and mice. In zebrafish, SorCS2 knockdown also affects neuromuscular junction morphology and fish motility. In mice, SorCS2 and PGRN are co-expressed by newborn MNs from embryonic day 9.5 until adulthood. Using cell-fate tracing and nerve segmentation, we find that SorCS2 deficiency perturbs cell-fate decisions of brachial MNs accompanied by innervation deficits of posterior nerves. Additionally, adult SorCS2 knockout mice display slower motor nerve regeneration. Interestingly, primitive macrophages express high levels of PGRN, and their interaction with SorCS2-positive motor axon is required during axon pathfinding. We further show that SorCS2 binds PGRN to control its secretion, signaling, and conversion into granulin. We propose that PGRN-SorCS2 signaling controls MN development and regeneration in vertebrates.

INTRODUCTION

Motor neurons (MNs) are the final relay transmitting signals from the central nervous system (CNS) to skeletal muscles, thereby enabling muscle contraction and movement. Their complex development requires a concerted action of trophic factors and guidance cues that determine cell-fate decisions, survival, axonal outgrowth, muscle targeting, and formation of functional neuromuscular junctions (NMJs).^{1,2} These processes rely on fine-tuned, spatiotemporal distribution of ligands, receptors, and transcription factors expressed by MNs themselves and by cells in close proximity, including macrophages or microglia. The same processes are often reactivated during axonal regeneration after nerve injury, supporting neuronal survival and axonal regrowth.^{3–5} Although MN diversification and motor axon outgrowth have been extensively studied, the cell communication and molecular mechanisms involved are not fully understood.

Murine spinal cord MNs develop from proliferating Olig2⁺ progenitors already found in the ventricular zone of ventral neural tube at embryonic day 9.5 (E9.5). They differentiate into postmitotic neurons by E10.5 and migrate laterally to initiate the anterior-posterior patterning. MNs mature in a stepwise manner.^{1,2} From E11, MNs start establishing six distinct anatomical structures called motor columns. While the lateral motor column (LMC) in brachial spinal cord innervates the forelimb, the neighboring medial motor column (MMC) innervates the epaxial muscle of the trunk.^{1,2,6} Nascent motor axons start innervating their target muscles between E11 and E13.5.^{1,2,7} Once they establish major axonal trajectories, MNs within each motor column initiate more refined differentiating and branching programs. They form diverse, muscle-specific clusters called motor pools, which is accompanied by terminal arborization.^{1,2} MN subclasses are distinguished by specific expression of transcription factors including Islet1/2, Foxp1, Pea3, and Homeobox proteins. Motor circuits fully mature between postnatal day 4 (P4) and P21.^{1,2,8}





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SorCS2 receptor is a member of the Vps10p-domain (D) family comprising sortilin, SorLA, SorCS1, and SorCS3. These single-pass transmembrane receptors predominate in neuronal populations of the central and peripheral nervous systems where they engage in cell sorting, signaling, and intercellular communication.^{9,10} They localize to vesicular compartments and the plasma membrane where they bind various ligands. This family has many roles in trafficking pathways that control distribution of synaptic proteins, ligand secretion, protein internalization, recycling, and lysosomal processing.¹⁰ Conceptually, signaling through a VPS10p-D receptor requires formation of heteromeric complexes with a ligand and a co-receptor.^{10–12} Notably, SorCS2 expression dynamically changes during embryogenesis, suggesting its key role in defining neuronal structures.^{13–16} SorCS2 controls various processes including growth-cone collapse, dopaminergic innervation, synaptic plasticity, and apoptosis.^{12,17–20} It is produced as a proform containing a propeptide, which is cleaved by furin in the *trans*-Golgi network (TGN) to generate a mature, single-chain receptor. The single chain can be converted into a two-chain variant, which is proapoptotic in peripheral glia.^{12,19} Importantly, SorCS2 binds secreted proneurotrophins pro-brain-derived neurotrophic factor (proBDNF) and pro-nerve growth factor (proNGF) to control axonal outgrowth, and it engages in synaptic targeting of tyrosine receptor kinase TrkB and *N*-methyl-D-aspartate (NMDA) receptors.^{19–21}

Progranulin (PGRN) is a multifunctional, secreted glycoprotein that predominates in microglia and neurons.²² It promotes neuronal development and repair, sustains neuronal integrity, and suppresses neuroinflammation.^{23–25} MN development is substantially perturbed in zebrafish lacking PGRN, while *Grn* (gene encoding PGRN) knockout mice display delayed regeneration of injured MNs.^{26–30} The neuroprotective effect of PGRN is further supported by *Grn* haploinsufficiency being causative for Alzheimer's disease^{31–34} and frontotemporal dementia (FTD).^{22,35–37} While insufficient PGRN levels associate with neurodegeneration, enhanced PGRN increases tumor growth, invasion, and mortality.^{38–40} PGRN exerts its pleiotropic functions by two modes of action. It signals across the plasma membrane and sustains lysosomal function via its cleavage into mature granulin peptides.^{23,24,30,41} How granulins operate in the lysosomes is unclear, but a role in ganglioside metabolism has been recently demonstrated.⁴² PGRN's complex functionality is likely executed by precise titration of PGRN outside and inside the cells. We previously identified sortilin as a critical receptor for internalization and lysosomal sorting of PGRN.^{43,44} Nevertheless, mechanisms involved in PGRN secretion and the receptor(s) responsible for its neurotrophic signaling remain unknown.

By combining biochemical approaches, transgenic models, advanced imaging, and image segmentation, we identify SorCS2 as a PGRN receptor that is required for MN diversification, NMJ morphology, and axon outgrowth in zebrafish and mice. We further demonstrate that SorCS2 binds PGRN to control its signaling, secretion, intracellular trafficking, and cleavage. These findings might be relevant for disorders associated with dysfunctional PGRN or SorCS2 function.

RESULTS

sorcs2 is expressed by spinal cord MNs in zebrafish embryos

To interrogate the function of SorCS2 in MN development, we first assessed its expression in zebrafish embryos. *sorcs2* transcripts were detected from 12 h post fertilization (hpf) with its expression gradually increasing during embryogenesis (Figure 1A). Data from single-cell transcriptome of zebrafish embryos⁴⁵ suggested enrichment of *sorcs2* RNA in spinal cord MNs and fast skeletal muscle cells at 24 hpf (Figures S1A–S1C), a time point of active MN development.^{46–48} We next analyzed *sorcs2* expression in whole-mount embryos at 24 hpf (Figures 1B–1E and S1D) and 48 hpf (Figures 1F, S1E, and S1F) using two RNA *in situ* hybridization (WISH) protocols⁴⁹ performed with the transgenic line *Tg(mnx1:GFP)^{ml2}* expressing GFP by *Mnx1*-positive MNs. By integrating Imaris-based surface-to-spot segmentation, we filtered for *sorcs2* transcripts localized inside the *Mnx1*⁺ neurons. Indeed, *sorcs2* was expressed by spinal cord MNs as well as in muscle cells at 24 hpf (Figures 1C–1E and S1D) and 48 hpf (Figures 1F, S1E, and S1F).

sorcs2 deficiency causes shorter and branched CaP motor axons

Zebrafish have three primary MN types innervating each somatic hemi-segment, where caudal primary (CaP) motor neuron grows first, extending its axon into the ventral somite.^{46–48} To study the impact of SorCS2 deficiency on MN outgrowth, we injected *sorcs2* splice-blocking morpholino (sMO) at 1-cell stage to mediate knockdown (KD). For rescue, we co-injected *hSORCS2* mRNA. The sMO perturbed *sorcs2* splicing at 24 hpf with no effect on developmental rate or morphology (Figures S2A–S2C). We then assessed motor axon length and branching in sMO-injected *mnx1:GFP* embryos at 27 hpf using Imaris-based filament segmentation (Figure 1G). KD embryos displayed stunted motor axons that are 40% shorter (Figure 1H) but have a higher number of branches (Figure 1I) and a larger total branch length (Figure 1J). These phenotypes were rescued with *hSORCS2* co-injection.

Figure 1. *sorcs2* is expressed by spinal cord MNs, and its loss causes shorter and more branched motor axons in zebrafish embryos

(A) RT-PCR shows *sorcs2* expression in zebrafish embryos.
(B) Zebrafish embryo at 24 hpf and analyzed areas.
(C) Alkaline phosphatase WISH of *sorcs2* at 24 hpf revealed *sorcs2* transcripts in somites and spinal cord (SC; arrows). Coronal section exhibits enriched *sorcs2* expression in the muscle (M) and notochord (NT).
(D–F) Fluorescent WISH labeled *sorcs2* transcripts in *Tg(mnx1:GFP)^{ml2}* zebrafish embryos. *sorcs2* was expressed by *Mnx1*⁺ MNs (arrow) at 24 hpf (D and E) and 48 hpf (F). (E and F) 3D view of z-stack confocal images of *sorcs2* transcripts in *mnx1:GFP* embryos. Unilateral sections were reconstructed using Imaris-based surface-to-spot segmentation and filtered for transcripts co-localized with *Mnx1*-GFP⁺ MNs.
(G–J) (G) Confocal z-stack images of injected *mnx1:GFP* embryos at 27 hpf. Motor axons were segmented as filaments (purple) and analyzed for their length (H), number of branches (I), and summarized (total) branch length (J). n = 63–70 neurons from 9–10 embryos/condition. All graphs show mean ± SEM. See also Figures S1 and S2.

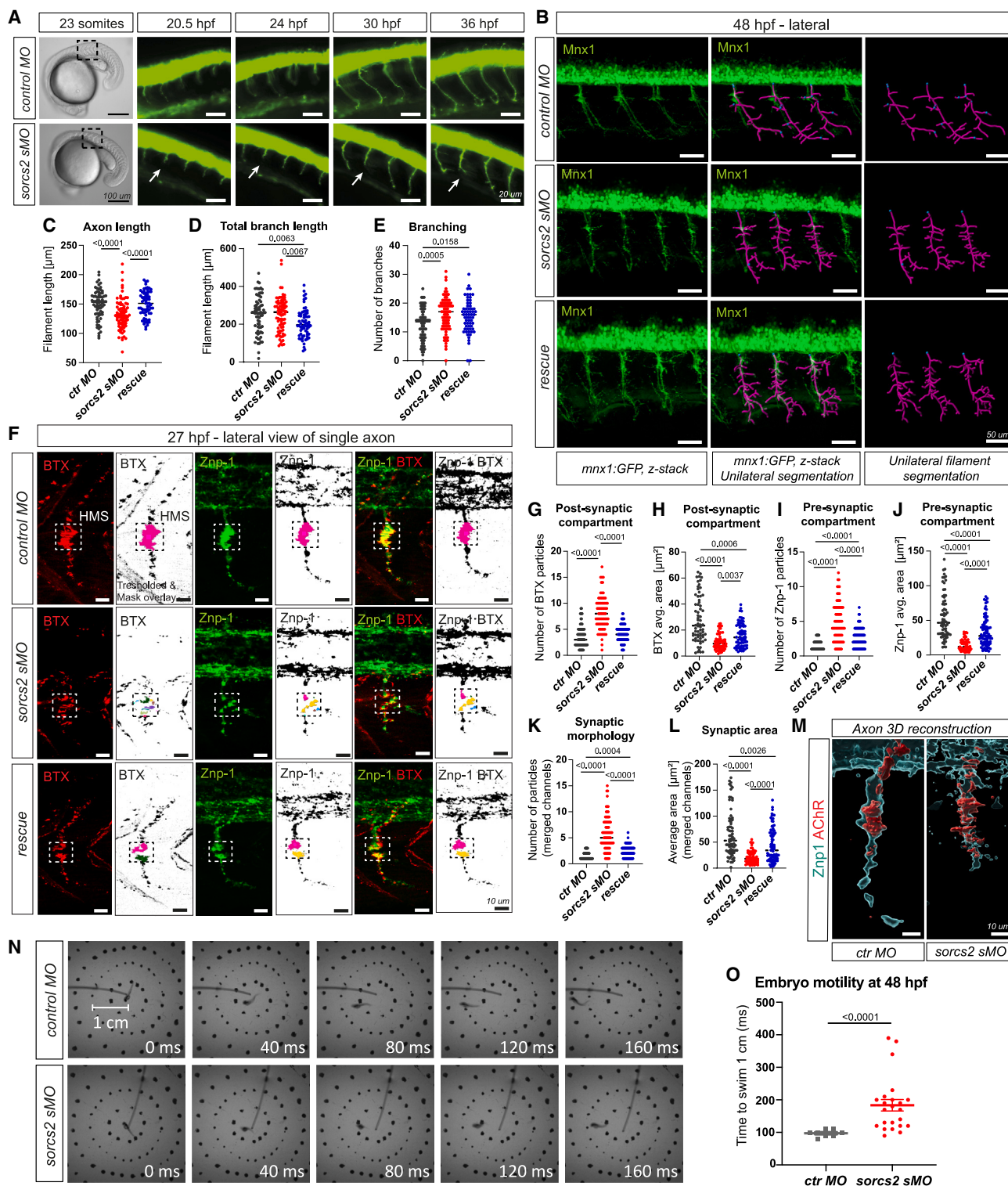


Figure 2. *sorcs2* deficiency causes morphological defects at NMJs and lower motility in zebrafish embryos

(A) Differential interference contrast images at 20.5 hpf and time-lapse of *Tg(mnx1:GFP)^{ml2}* embryos from 20.5 to 36 hpf. Truncated motor axons (arrows) of *sorcs2* KD embryos pause longer at horizontal myoseptum (HMS) before continuing ventrally, but remain truncated at later stages.

(B–E) (B) Confocal z-stack images of *mnx1:GFP* embryos at 48 hpf with reconstructed motor axons (purple). Filament quantification of axon length (C), branching (E), and total branch length (D). $n = 70$ –84 neurons from 10–12 embryos/condition.

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To determine motor axon outgrowth in non-transgenic animals, we injected *wild-type* (*WT*) fish and performed whole-mount immunofluorescence (IF) for synaptotagmin-2 using Znp-1 antibody and scored axon outgrowth through the myotome (Figures S2D and S2E). In control embryos, 95.6% of the CaP axons reached the distal ventral myotome, and all axons extended beyond the notochord. In contrast, axons of *sorcs2* KD embryos were stunted, with 81.9% of axons reaching the distal ventral myotome and only 1.9% reaching the horizontal myoseptum (Figure S2F). As expected, 16.5% of the axons were branched in *sorcs2* KD compared to 5.9% in control (Figure S2G). Both phenotypes were rescued with *hSORCS2* co-injection. We further analyzed MN length in *Sorcs2* knockout (*KO*) embryos from heterozygous cross of *sorcs2*^{sa11642} line. Similar to *sorcs2* KD, *KO* embryos exhibited stunted CaP axons (Figure S2H). sMO injection into *Sorcs2* *KO* embryos did not aggravate this phenotype (Figure S2I), confirming specificity of the morpholino. Owing to poor breeding of the *sorcs2*^{sa11642} line, we continued using the KD model.

To explore the dynamics of motor axon outgrowth in *sorcs2* KD embryos, we subjected *mnx1:GFP* embryos to time-lapse imaging. CaP axons exit the spinal cord around 16–18 hpf, and reach the horizontal myoseptum around 20–24 hpf where they make first synaptic contacts. They pause here for a few hours before continuing ventrally into the myotome.⁵⁰ *sorcs2* KD axons paused for a longer period at the horizontal myoseptum, although they later extended beyond the choice point and grew ventrally (Figure 2A; Videos S1 and S2). However, several axons remained truncated and some started branching prematurely at later stages. We thus quantified *Mnx1*⁺ axon length and branching at 48 hpf using Imaris-based filament segmentation (Figures 2B–2E). We found that KD embryos still display shorter motor axons at this stage (Figure 2C) with a higher number of branches (Figure 2E), although the total branch length no longer differs (Figure 2D).

Taken together, these data indicate that *SorCS2* deficiency during early embryogenesis impairs motor axon outgrowth, pathfinding, and branching of zebrafish spinal cord MNs.

***sorcs2* deficiency causes defects in neuromuscular synapse and reduced motility of zebrafish embryos**

Besides being an important decision point for axon branching, the horizontal myoseptum is the first and one of the major synaptic sites for motor axons to connect with muscle pioneers. Importantly, the growth of new axon branches originates and depends on the synapse stabilization at the horizontal myoseptum.^{51,52} Given the aberrant motor axon outgrowth and branching in *sorcs2*-deficient embryos, we addressed the role of *sorcs2* during synaptogenesis. We labeled presynaptic compartments with Znp-1 and postsynaptic compartments with α -bungarotoxin

(BTX) at 27 hpf. We then quantified single axons at the horizontal myoseptum in terms of synapse morphology according to Opris, oreanu et al.⁵¹ using thresholding and masking (Figures 2F–2M and S2J). *sorcs2* KD fish exhibited severely fractioned synapse morphology (Figures 2K and 2M) with smaller average synaptic area (Figure 2L), both at postsynaptic (Figures 2G and 2H) and presynaptic (Figures 2I and 2J) compartments. Impaired synaptogenesis was vastly rescued with *hSORCS2* co-injection. Because of high synaptic complexity at 48 hpf,⁵² we did not quantify the neuromuscular junction; however, the synapse morphology of KD embryos still differed from control embryos at this stage (Figure S2K). We next investigated embryo motility at 48 hpf. We performed a touch-response assay⁵³ that assesses the time spent to swim 1 cm after a light touch. *sorcs2* KD embryos showed a slower escape response as they spent ~183 ms to swim 1 cm compared to the controls that needed ~98 ms (Figures 2N and 2O; Videos S3 and S4).

Our results demonstrate that *SorCS2* loss during early synaptogenesis causes anatomical and functional defects at the major synaptic point of NMJs, which likely translates into movement deficits.

***SorCS2* is expressed by murine MNs, and its loss impairs axonal outgrowth**

Since *SorCS2* is conserved among vertebrates,^{10,13,54,55} we explored *SorCS2* function in murine spinal cord. We first examined *SorCS2* protein levels and a number of markers (Table S1). At E12.5, *SorCS2* was expressed by Hb9⁺ MNs including limb-innervating *Foxp1*⁺ LMC. *SorCS2* immunoreactivity exhibited a comparable pattern in brachial (Figures 3A and 3B), thoracic (Figure S3A), and lumbar spinal cord (Figure S3B). Additionally, *SorCS2* was highly expressed in the floor plate and roof plate (Figures 3A and 3B) while being detected in low-to-zero levels in the dorsal spinal cord (Figure 3A). *SorCS2* immunoreactivity in dorsal root ganglion (DRG) was overall low compared to ventral horn (Figures 3A–3C and S12D). Nevertheless, some *Islet1*⁺ sensory neurons exhibited higher levels of *SorCS2* while their *Sox10*⁺ progenitors had low-to-zero *SorCS2* (Figure S12D). At this age, *SorCS2* was also highly expressed in skeletal muscle, esophagus, trachea, and Schwann cell precursors (Figures 3A and S12D). When exploring the earlier stages of the brachial spinal cord, *SorCS2* was detected in *Olig2*⁺ motor neuron progenitors at E9.5, with its signal increased in the newborn MNs (Figure S3C). We observed the same pattern at E10.5, E11.5, and E12.5 (Figures S3D–S3F). *SorCS2* was highly expressed by the floor plate at all analyzed stages. Next, we examined whether all MNs in brachial spinal cord express *SorCS2*. We combined *Islet1* and *Foxp1* staining to distinguish lateral motor columns (LMC: LMCM, medial [*Islet*⁺, *Foxp1*⁺]; LMCL, lateral [*Islet*⁺, *Foxp1*⁺]) from the medial motor column

(F) IF staining of the NMJs show z-stack projections of single axons labeled by Znp-1 (synaptotagmin-2; presynaptic compartment) and BTX-555 (AChRs; postsynaptic compartment) at 27 hpf. Black-and-white panels show segmented images with quantified NMJs.

(G–L) Quantification of HMS area revealed fragmented synapse morphology (K) while covering smaller synaptic area (L) in postsynaptic (G and H) and presynaptic (I and J) compartments. n = 90–99 axons from 10–11 embryos/condition.

(M) Axonal 3D reconstruction of (F).

(N) Touch-response assay of 48 hpf embryos.

(O) Quantification of embryo motility. n = 10–21 embryos/genotype.

All graphs show mean $\pm$ SEM. See also Figure S2; Videos S1, S2, S3, and S4.

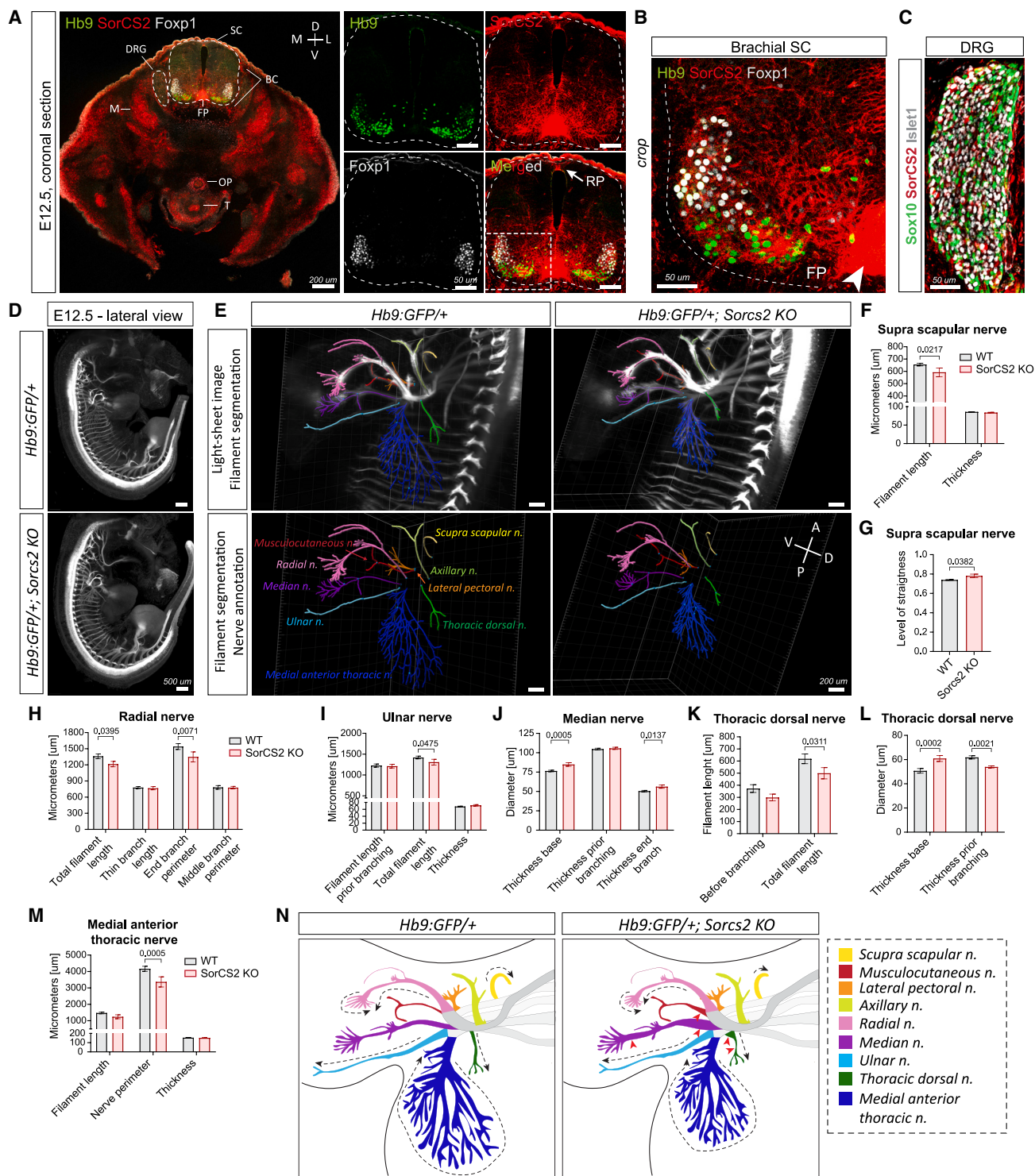


Figure 3. SorCS2 is expressed by murine MNs, and its loss causes shorter and thicker brachial nerves

(A–C) IF staining of SorCS2 and MN markers in E12.5 mouse embryos. (A) SorCS2 is highly expressed in ventral horn, floor plate (FP; arrowhead), roof plate (RP; arrow), muscle (M), esophagus (OP), trachea (T), and boundary cap (BC) cells. D, dorsal; V, ventral; M, medial; L, lateral. (B) SorCS2 is highly expressed by all brachial MNs, in contrast to DRG precursors (Sox10⁺) and sensory neurons (Islet1⁺) (A and C).

(D and E) Light-sheet images of whole-mount E12.5 embryos from *Hb9:GFP* lines alone (D), and combined with Imaris-based filament segmentation of nine annotated⁸ brachial nerves (E). D, dorsal; V, ventral; A, anterior; P, posterior.

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(MMC: Islet⁺, Foxp1⁺). Indeed, all three motor columns already express SorCS2 at E11.5 but at different levels, with lowest signal observed in LMCm and highest in MMC (Figures S3G and S3H). Pea3-positive MNs that distinguish the LMC motor pool innervating cutaneous maximus and latissimus dorsi muscles also expressed SorCS2 (Figures S3I and S3J).

Given the consistent SorCS2 expression in all MNs, we asked whether SorCS2-deficient mice, like zebrafish, display innervation deficits. Hence, we collected E12.5 embryos from *Hb9:GFP/+* (heterozygotes) and *Hb9:GFP/+; Sorcs2^{-/-}* transgenic lines to perform whole-mount light-sheet microscopy and examine GFP⁺ motor nerves originating from brachial, thoracic, and lumbar spinal cord (Figure 3D; Videos S5, S6, S7, and S8). This strategy allowed very precise and efficient 3D annotation and segmentation of large specimens. We segmented nine forelimb/brachial nerves^{8,56} originating from C4–T1 spinal segments (i.e., brachial spinal cord) (Figure 3E and Video S9), seven thoracic nerves of T6–T12 spinal segments (Figure S4O), and three hindlimb nerves⁵⁷ (Figure S4H). In the forelimb, SorCS2 deficiency leads to a shorter and straighter supra-scapular nerve (Figures 3F and 3G), a shorter radial nerve exhibiting also smaller end branching (Figure 3H), a shorter ulnar nerve (Figure 3I), thicker median (Figure 3J) and musculocutaneous nerves (Figure S4C), a shorter thoracic dorsal nerve with varying thickness along its length (Figures 3K and 3L), and a smaller medial anterior thoracic nerve (Figure 3M). Other brachial nerves and/or additional morphological features were unchanged (Figures S4A–S4G). The data are summarized in Figure 3N. We observed no difference in T6–T12 thoracic nerves (Figures S4P–S4W) and hindlimb nerves (Figures S4I, S4J, S4L, and S4M) except for the deep peroneal nerve that was shorter (Figure S4K).

Regardless of its expression in all murine MNs, the loss of SorCS2 affects the development of specific nerves, mostly those of the C4–T1 spinal segments, resulting in formation of shorter and/or thicker nerves.

SorCS2 controls cell-fate decisions of brachial MNs in the lateral motor column

Impairments in membrane receptors during MN development often influence not only motor axon pathfinding but also MN refined differentiation required for muscle-specific targeting.^{8,58,59} Therefore, we quantified the number of MNs in brachial motor columns in *WT* and *Sorcs2 KO* E12.5 embryos using Foxp1 and Islet1 staining (Figure 3). We histologically distinguished the quantifiable area with Pea3 expression, which is restricted to C4–T1 spinal segments^{60,61} (Figures 4A, 4B, S5A, and S5B). Since Pea3 controls the terminal arborization of thoracic dorsal and medial anterior thoracic nerves,⁶⁰ we also quantified the corresponding motor pool. Our cell-fate tracing revealed that SorCS2 deficiency causes elevated MN number (Figure 4C). This difference originated specifically from an increased MN count in the LMCI subpopulation (Figures 4D and 4F) as

LMCm and MMC remained unchanged (Figures 4E and 4G). Interestingly, Pea3⁺ motor pool was not affected (Figure 4H). We then addressed whether the changes in cell-fate decision correspond to the topographical position of MN clusters forming specific nerves⁸ (Figures 4I and S5). We compared single sections and found a significant increase in posterior LMCI (Figure 4J), which roughly corresponded to C7–T1 spinal segments (Figure 4I). These sections contain MN somas that form radial, median, ulnar, thoracic dorsal, and medial anterior thoracic nerves⁸ (Figure 4I), thus exactly those nerves exhibiting impaired outgrowth due to *Sorcs2* loss (Figures 3E–3N). No changes were seen in single sections of LMCm, MMC, and Pea3⁺ populations (Figures 4K–4M). These data demonstrate that SorCS2 is not only required for axonal outgrowth of specific MN clusters but also regulates their cell-fate decisions.

SorCS2 binds PGRN, and PGRN-expressing primitive macrophages interact with motor axons

Since SorCS2 is a membrane receptor that controls axonal outgrowth, we tested a panel of molecules implicated in axon pathfinding and trophic support using surface plasmon resonance. The panel included Ephrins, PGRN, Netrin-1, fibronectin leucine-rich repeat transmembrane proteins, and Netrin receptor UNC5D^{2,23,62–64} (Table S2). The majority of tested proteins showed no binding to the immobilized extracellular domain of SorCS2. We detected weak binding only for EphrinA1–4, with inconsistent results between mouse and human proteins. In marked contrast, we found a very strong interaction with PGRN (Figure 5F and Table S2). This was intriguing, since KD of the PGRN-encoding gene *grna* in zebrafish phenocopies truncation defects and premature branching of MNs^{26–28} observed in the *sorcs2* KD embryos (Figures 1 and 2). To test whether SorCS2 functionally interacts with PGRN *in vivo*, we first explored PGRN expression during embryogenesis. Data from zebrafish single-cell transcriptome⁴⁵ revealed that *grna* RNA was enriched in embryonic macrophages and neutrophils (Figures S6A–S6C). We then performed fluorescent WISH against *grna* using *mnx1:GFP* fish at 24 hpf, 27 hpf, and 48 hpf (Figures 5A–5C, S6D, and S6E). Even though the signal was rather diffused at 24 hpf and 27 hpf, it was enriched in cells of ventral trunk (Figures S6D and S6E), likely corresponding to immune cells.^{65,66} Interestingly, we observed sparse yet distinct signal surrounding the *Mnx1*⁺ axons, especially at 27 hpf (Figures 5A and 5B). This stage corresponds to E10.5 of mouse development.⁶⁵ Spot-to-surface segmentation further identified *grna* transcripts inside the MNs, although the fluorescence was very weak (Figures S6D and S6E). Strikingly, when we analyzed 48 hpf embryos (corresponding to E14.5), we observed high *grna* expression in cells closely associated with motor axons, which was confirmed by spot-to-surface segmentation and filtering for maximum 5 μ m distance from *Mnx1*-positive surface (Figure 5C). These cells likely account for primitive macrophages that migrated

(F–M) Quantification of the brachial nerves. (F and G) Supra-scapular nerve is shorter and straighter. (H) Radial nerve is shorter and its end branch perimeter is smaller while the side branches are unchanged. (I) Ulnar nerve is shorter while its thickness is unchanged. (J) Median nerve is thicker at its base but also before the final branching. (K) Thoracic dorsal nerve is shorter and thicker at its base (L), but narrower before the branching (L). (M) Medial anterior thoracic nerve perimeter is smaller while the thickness is unchanged.

(N) A scheme summarizing observed phenotypes. Dashed arrows, affected length; arrowheads, changed thickness.

n = 9 embryos/genotype. All graphs show mean $\pm$ SEM. See also Figures S3, S4, and S12; Videos S5, S6, S7, S8, and S9.

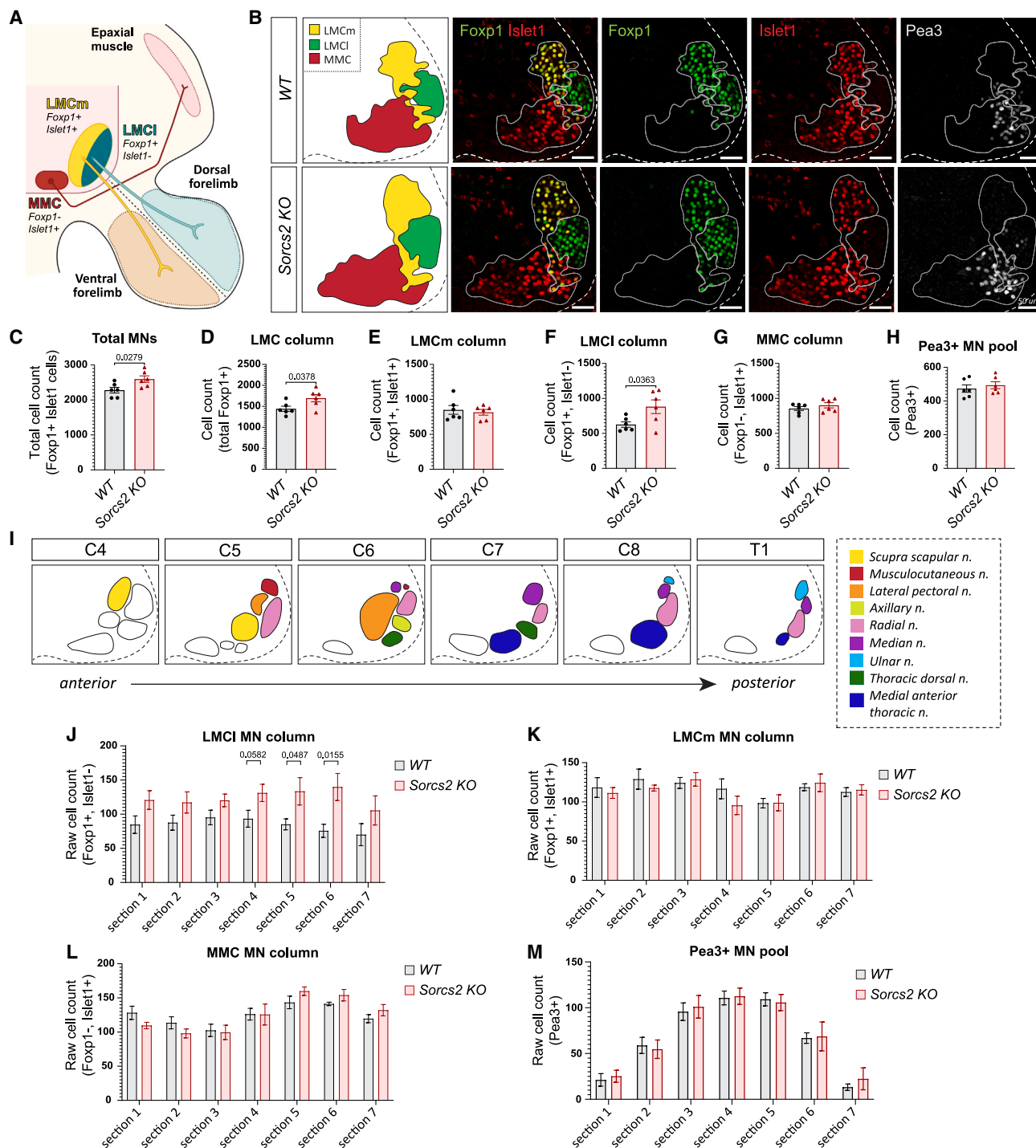
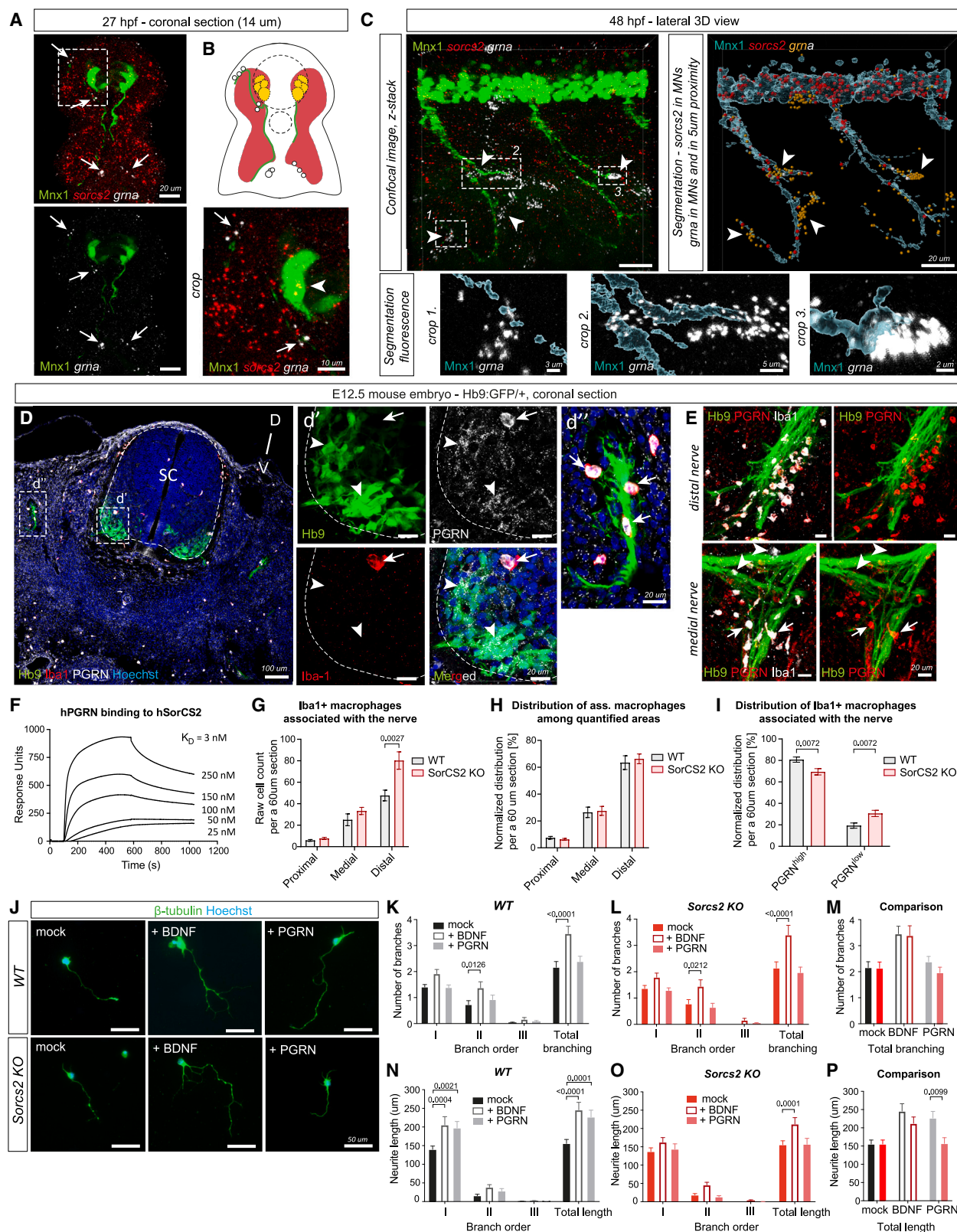


Figure 4. SorCS2 loss impairs cell-fate decisions of brachial MNs in lateral motor column

(A) Motor columns of brachial spinal cord, their transcription factors, and their target areas at E12.5. (B) Coronal sections of right ventral horn at E12.5 with the corresponding motor columns labeled by IF. (C–H, J–M) Quantification of the MN cell-fate tracing. The cell count was either summarized (C–H) or presented in the individual sections for anterior-posterior occurrence (J–M). (C) *sorcs2* KO embryos exhibit higher MN count, originating from increased LMC number (D), specifically the LMCI population (F and J). LMCm (E and K) and MMC (G and L) clusters were unaltered. Pea3+ motor pool did not change (H and M). The increase in LMCI MN count was significant in the posterior sections (J), which contain MNs of radial, medial, ulnar, medial anterior thoracic, and thoracic dorsal nerves. (I) Topographical localization of brachial MNs at E12.5, color-coded according to brachial nerves they form.⁸ n = 7 sections/embryo, 6 embryos/genotype. All graphs show mean ± SEM. See also Figure S5.



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into the tissue from the yolk sac.^{65,66} We detected low-to-zero *grna* in the MNs at this stage.

We next investigated *Gm* (gene encoding PGRN) expression in murine MNs. Data from single-cell RNA sequencing of embryonic spinal cord⁶⁷ indicated that *Gm* and *Sorcs2* are co-expressed by a number of neuronal populations from E9.5 to E13.5 (Figures S7A and S7B; Table S3), including MNs and their progenitors (Figures S7B and S7C). These data were confirmed by RNA *in situ* hybridization (Figure S7D). We next evaluated PGRN protein levels. Identically to SorCS2, vesicular PGRN was already detected in MN progenitors with increased signal in the newborn MNs at E9.5. This pattern was maintained also at E10.5, E11.5, and E12.5 (Figures 5D and S8A–S8D). Similarly, PGRN was already detected in all motor columns in brachial spinal cord at E11.5 (Figures S8E and S8F), as well as in Pea3⁺ motor pool (Figures S8G and S8H). At E9.5 and E10.5, PGRN was mostly detected in the floor plate and notochord (Figures S9A and S9B). However, at E10.5, we noticed cells outside the spinal cord that were highly positive for PGRN, with their number gradually increasing at E11.5 and E12.5, respectively (Figures S9C and S9D). Like in zebrafish, these cells were Iba1⁺ primitive macrophages and microglia (Figure 5D) that have migrated into the trunk from the yolk sac.^{65,66} Strikingly, these macrophages were physically associated with the Hb9:GFP-positive motor axons, especially medially and distally from the ventral horn and at branching points (Figures 5E and S9D), as seen in zebrafish (Figure 5C). This was shown only for sensory neurons.⁶⁸ We detected low-to-zero PGRN in Sox10⁺ migrating neural crest cells, Sox10⁺ DRG precursors, Sox10⁺ Schwann cell precursors, and Islet1⁺ DRG sensory neurons (Figures S10A–S10C).

Altogether, our data suggest that PGRN inside the motor neurons or secreted by cells surrounding the growing axon may selectively regulate MN development through binding to SorCS2. This hypothesis could explain why SorCS2 loss of function does not affect all nerves equally.

PGRN-induced motor neurite outgrowth requires SorCS2 expression in MNs

Macrophage interaction with the motor nerve was already apparent at E11.5 (Figure S10C). We thus addressed whether SorCS2 deficiency affects the number and distribution of motor-nerve-associated macrophages at E12.5. We used 60- μ m-thick sections of brachial spinal cord from our Hb9:GFP/+ lines. We

then quantified Iba1⁺ primitive macrophages associated with the Hb9:GFP signal located proximally (until first branching), medially (until notochord), and distally from the ventral horn (Figures S9D and S9E). Strikingly, SorCS2 loss caused an increased number of macrophages at the distal nerve (Figure 5G), while their distribution within the sections was unchanged (Figure 5H). We observed that 80% of WT macrophages have high levels of PGRN (PGRN^{high}), while 20% show low-to-zero PGRN (PGRN^{low}) (Figures 5E and 5I). Remarkably, *Sorcs2* KO embryos displayed a 10% decline in PGRN^{high} macrophages translating into an increased number of PGRN^{low} macrophages (Figure 5I). These observations indicate compensatory upregulation of PGRN release by macrophages into the extracellular space toward the motor axons lacking SorCS2 receptor. Therefore, SorCS2 expression in MNs might be required for PGRN-induced MN outgrowth.

To test this hypothesis, we compared neurite outgrowth and sprouting in primary MNs isolated from E13.5⁶⁹ WT and *Sorcs2* KO spinal cords and cultured them for 4 days in the absence or presence of exogenous PGRN. We used recombinant BDNF as a positive control, and no stimulation (mock) as negative control (Figure 5J). There was no effect on branching in WT or KO neurons following PGRN stimulation (Figures 5K–5M). However, PGRN treatment substantially increased total neurite length by 46% in the WT (Figure 5N), while *Sorcs2* KO neurons were completely refractory to PGRN stimulation (Figures 5O and 5P). This was not accounted for by an intrinsic failure of the neurons to respond to trophic stimulation, since addition of BDNF increased both branching and neurite length in the *Sorcs2* KO and WT cultures to a similar extent (Figures 5M and 5P).

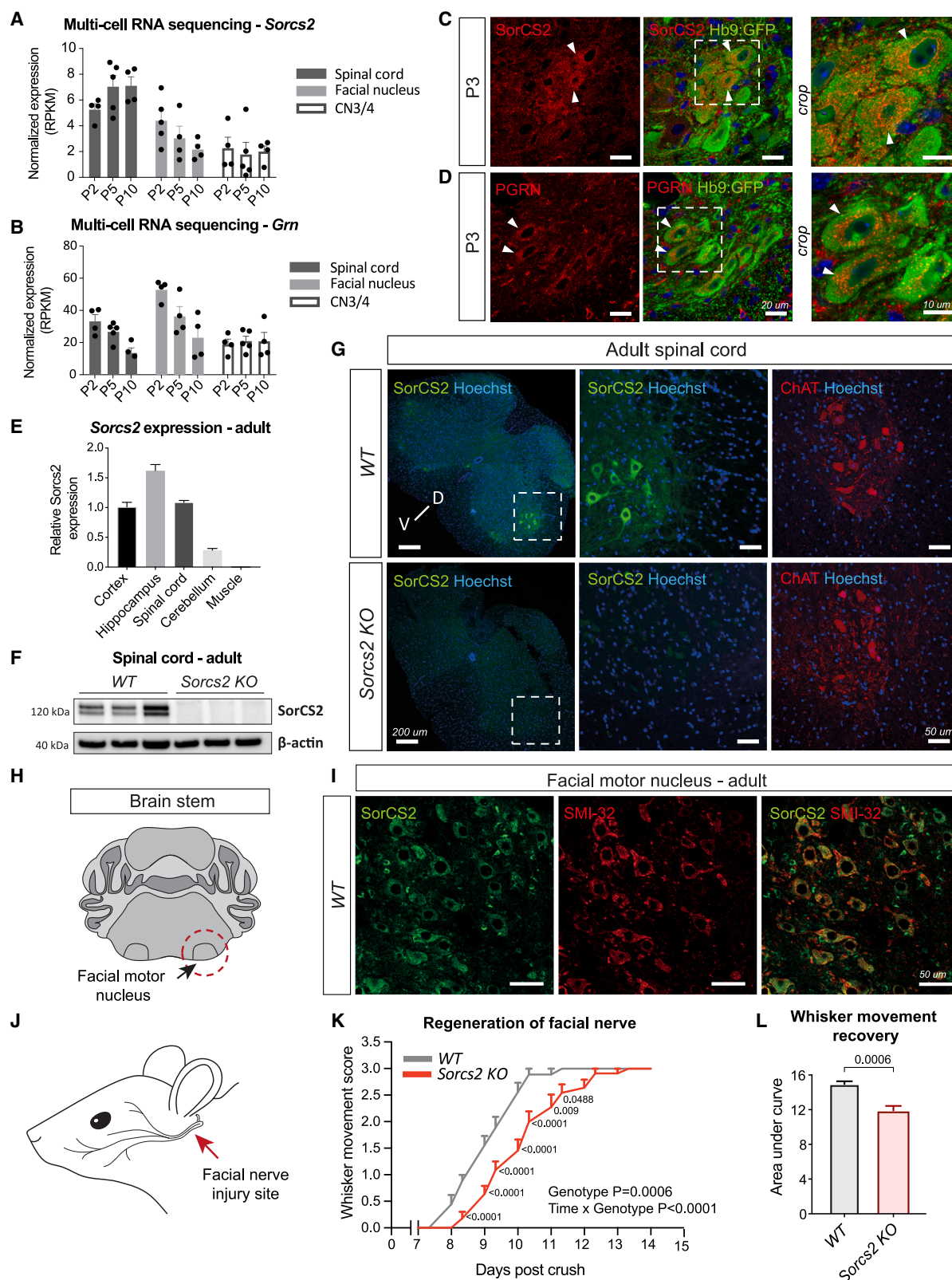
Using *in vitro* and *in vivo* systems, we thus demonstrate that SorCS2 likely engages in macrophage-to-neuron cell communication during motor nerve outgrowth as a receptor for secreted PGRN.

SorCS2 and PGRN are co-expressed by postnatal MNs

During postnatal MN development, the dendritic tree grows to attain maturity and establishes fully functioning synapses.⁷⁰ PGRN is neuroprotective in adult mice and supports neuronal function.^{44,71} Hence, we explored SorCS2 and PGRN expression in postnatal MNs. We extracted multi-cell RNA sequencing data from laser-captured, micro-dissected murine MNs⁷² and found that *Sorcs2* (Figure 6A) and *Gm* (Figure 6B) are co-expressed

Figure 5. PGRN-induced motor neurite outgrowth requires SorCS2 expression in MNs

(A) HCR3.0 on *Tg(mnx1:GFP)^{ml2}* embryos shows enriched *grna* in close proximity to motor axons (arrows) at 27 hpf, while *sorcs2* is expressed by MNs (arrowhead). (B) Illustration depicting *sorcs2* expression by MNs (yellow) and skeletal muscle (red), while *grna* (white) is enriched along motor axons (green). (C) HCR3.0 of *grna* and *sorcs2* in *Tg(mnx1:GFP)^{ml2}* embryos at 48 hpf followed by Imaris-based segmentation. *grna* transcripts were filtered in cells within 5 μ m proximity to *Mnx1*⁺ surface (orange; arrowheads) and inside the MNs (white). (D and E) (D) IF of PGRN and Iba-1 in coronal sections of brachial spinal cord of *Hb9:GFP/+* embryos at E12.5. Hb9⁺ MNs express PGRN (arrowheads). PGRN is highly expressed by Iba1⁺ microglia (d') and macrophages (d'') (arrows). (E) PGRN⁺ macrophages at branching points of medial and distal nerves. Some nerve-associated macrophages have low PGRN (arrowheads). SC, spinal cord; D, dorsal; V, ventral. (F) Surface plasmon resonance revealed concentration-dependent binding of hPGRN to immobilized hSorCS2. (G–I) Quantification of PGRN⁺ macrophages associated with the Hb9⁺ motor axons in brachial spinal cord at E12.5. (G) *Sorcs2* KO embryos exhibit more macrophages associated with the distal motor nerve. (H) Distribution of associated macrophages among quantified areas. (I) Distribution of nerve-associated macrophages in relation to their intracellular PGRN levels. n = 3 sections/embryo, 4 embryos/genotype. (J) IF of primary MN cultures grown without stimulation (mock), with 5 ng/mL BDNF, or with 100 ng/mL PGRN. (K–P) Quantification of primary motor axon outgrowth as number of branches (K–M) and neurite length (N–P) of primary, secondary, and tertiary neurites, and when summarized. WT: mock = 37 neurons (n.), BDNF = 37 n., PGRN = 39 n. *Sorcs2* KO: mock = 36 n., BDNF = 38 n., PGRN = 31 n. All graphs show mean $\pm$ SEM. See also Figures S6–S10 and Table S2.



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by MNs in the spinal cord, the facial nucleus, and the cranial nuclei 3 and 4 at P2, P5, and P10. Therefore, *Sorcs2* and *Gm* co-expression is not restricted only to spinal cord MNs during early embryogenesis. These data were validated on protein levels in *Hb9:GFP/+* pups at P3. We observed that SorCS2 and PGRN are abundant in Hb9-positive neurons in the spinal cord, where they localized predominantly in vesicular compartments (Figures 6C and 6D). Similarly, qRT-PCR revealed that *Sorcs2* mRNA transcripts are prominent in adult spinal cord at levels comparable to those in cortex (Figure 6E). We confirmed the presence of SorCS2 protein in adult spinal cord by immunoblotting (Figure 6F) and IF (Figure 6G), which showed that SorCS2 was restricted to ChAT⁺ MNs. Similarly, *Gm* transcripts and PGRN protein were detected in the adult spinal cord MNs (Figures S11A–S11C), an observation also reported by others.²⁴

Adult *Sorcs2* KO mice show delayed regeneration of the facial nerve

Gm KO adult mice exhibit delayed recovery after a facial nerve injury,³⁰ suggesting that PGRN supports axonal outgrowth. The facial nerve, which is required to move the whiskers, is formed by motor axons originating from the facial motor nucleus in the brain stem (Figures 6H–6J). IF using SMI-32 that labels neurofilament H showed that SorCS2 was present in facial motor nucleus MNs (Figure 6I), previously also reported to express PGRN.³⁰ We therefore investigated regeneration capacity of the facial nerve of *Sorcs2* KO mice following facial nerve injury.³⁰ The facial nerve was gently crushed distally to the retroauricular branch on one side. The contralateral (uninjured) side was used as a control (Figure 6J). Regeneration was subsequently assessed by scoring provoked whisker movement twice a day. Injured motor axons resulted in immobile whiskers for over 7 days (Figure 6K). While *WT* mice fully recovered within 12 days, *Sorcs2* KO mice required 14 days. Quantified as the area under the curve, we show that *Sorcs2* KO display significantly delayed whisker-movement recovery (Figure 6L). Remarkably, the delay fully phenocopied that observed in *Gm* KO mice.³⁰

Our data demonstrate that SorCS2 is involved in motor nerve regeneration and suggest that SorCS2 can functionally interact with PGRN when co-expressed in adult MNs.

Overexpressed SorCS2 retains PGRN intracellularly and reduces its secretion

Given that the extracellular domain of SorCS2 binds ectopically added PGRN (Figure 5F) and that both genes are co-expressed by pre- and postnatal MNs, we further explored the interaction be-

tween SorCS2 and PGRN. To this end, we first co-cultured HEK293 cells transfected with either hPGRN or hSorCS2, reasoning that secreted PGRN from cell 1 would bind to SorCS2 receptor at the plasma membrane of cell 2 (*trans* interaction). Surprisingly, we failed to detect any *trans* interaction in co-cultures whether analyzed by IF (Figures 7A, S11D, and S11E) or co-immunoprecipitation (co-IP) (Figure 7C). Similar results were obtained with N-terminally HA-tagged PGRN, which excludes the possibility that anti-PGRN antibody interferes with PGRN binding to SorCS2 receptor due to steric hindrance (Figures S11F, S11G, and S11J). These results follow other studies demonstrating that SorCS2 requires a co-receptor to bind its extracellular ligands.¹²

Aside from facilitating formation of signaling receptor complexes,^{12,17,18} SorCS2 also mediates endocytosis and anterograde and retrograde trafficking in neurons.^{12,19,21,73} To assess whether SorCS2 and PGRN form a complex intracellularly (*cis* interaction), HEK293 cells were co-transfected to overexpress both proteins. We found that SorCS2 and PGRN strongly co-localized in a perinuclear compartment and vesicular structures, likely representing TGN, endosomes, and sorting vesicles (Figure 7B and S11H). We also stained endogenous PGRN in SorCS2-expressing cells. Localization of endogenous PGRN appeared identical to the overexpressed PGRN/PGRN-HA. Moreover, endogenous PGRN also co-localized with hSorCS2 in vesicular structures (Figure S11I). To demonstrate a physical interaction between the proteins in *cis*, we performed co-IP using untagged (Figure 7C) and HA-tagged PGRN (Figure S11J). We indeed observed strong interaction of intracellular PGRN with SorCS2 using both strategies. This interaction was restricted to the SorCS2 proform that is enriched in the TGN. Notably, hSorCS2 overexpression resulted in a ~300% increase in intracellular PGRN (Figures 7C, 7E, and S11J), which translated to a ~75% decrease in PGRN secretion (Figures 7D, 7F, and S11K). The total amount of PGRN produced by the cells (i.e., sum of PGRN in the medium and lysate) was decreased (Figure 7G), suggesting that SorCS2 may control PGRN secretion, PGRN trafficking from the TGN to lysosomes for degradation, and/or PGRN cleavage into granulins.

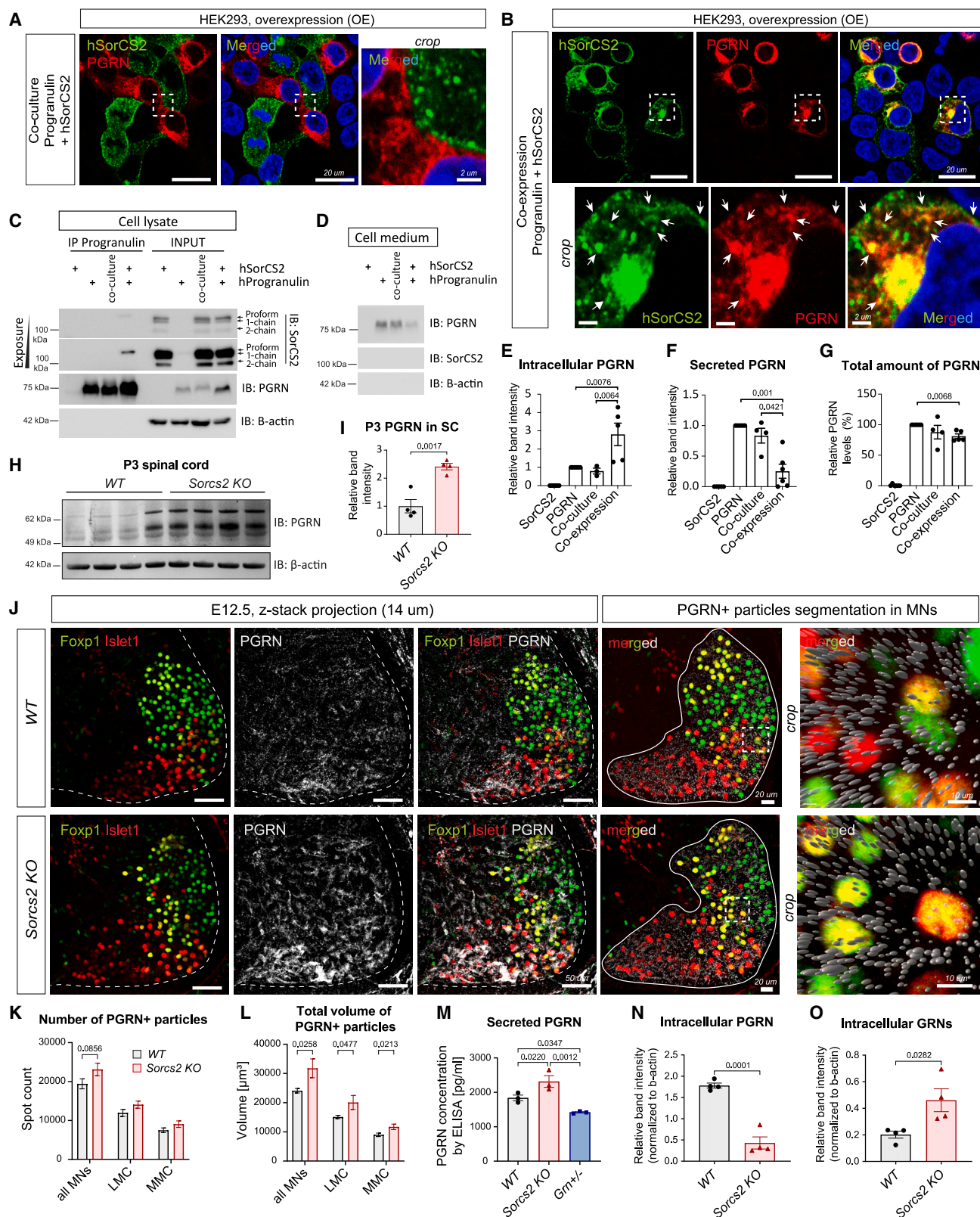
We demonstrate that SorCS2 has the capacity to bind and retain PGRN intracellularly when co-expressed in the same cells by which it regulates PGRN trafficking and secretion.

SorCS2 loss of function increases PGRN secretion and the production of granulins

We next assessed whether SorCS2 loss alters PGRN levels *in vivo*. In accordance with reduced PGRN levels in

Figure 6. Adult *Sorcs2* KO mice show delayed functional recovery after facial nerve crush injury

(A and B) Multi-cell RNA sequencing of laser-captured MNs⁷² show *Sorcs2* (A) and *Gm* (B) expression in postnatal MNs.
(C and D) IF of ventral spinal cord (coronal section) confirmed SorCS2 (C) and PGRN localization (D) in postnatal Hb9⁺ MNs (arrowheads). Nuclei labeled with Hoechst.
(E) *Sorcs2* qRT-PCR in 12-week-old *WT* mice (n = 3). Relative expression indicates fold change normalized to cortical expression ($2^{-\Delta\Delta Ct}$).
(F and G) Immunoblotting of adult spinal cord shows SorCS2 in *WT* but not in *Sorcs2* KO samples (F), which was confirmed by IF (G). β -Actin served as a loading control.
(H) Localization of facial motor nucleus.
(I) IF of SorCS2 and the neuronal marker SMI-32 in facial motor nucleus.
(J) Scheme of the facial nerve and the injury site.
(K) Regeneration of the facial nerve, scored on the basis of whisker movement. n = 9–11.
(L) Area under the curve of whisker-movement recovery scores.
All graphs show mean $\pm$ SEM. See also Figure S11.



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hSorCS2-overexpressing cells, PGRN was increased up to 240% in the spinal cord of *Sorcs2* KO pups at P3 (Figures 7H and 7I) despite comparable mRNA levels (Figure S11M). The PGRN increase was confirmed using another antibody and cohort of animals (Figure S11L). To test whether elevation of PGRN in SorCS2-deficient tissue already appears in embryonic MNs, we stained PGRN in the MN domain of E12.5 brachial spinal cord (Figure 7J, left panels). We then segmented the PGRN signal as spots of differential sizes (Figure 7J, right panel; Video S10) and analyzed the number, volume, and distribution of PGRN⁺ particles in LMC and MMC motor columns. *Sorcs2* KO MNs displayed a non-significant yet increasing trend in PGRN⁺ particle number (Figure 7K). Indeed, the total volume of PGRN⁺ particles per section was significantly increased in both motor columns and in the MN domain of *Sorcs2* KO embryos (Figure 7L). Distribution of PGRN⁺ particles between the motor columns did not change, nor did the average volume of a single PGRN⁺ particle (Figures S12A and S12B).

Since this approach could not distinguish between extracellular and intracellular PGRN nor granulins, we took advantage of a commercial ELISA kit to determine whether SorCS2 deficiency affects PGRN release in cultured neurons. We analyzed media from WT and *Sorcs2* KO primary cortical cultures that were prepared from P0 pups and cultivated them for 7 consecutive days. We used *Gm*^{+/-} pups as a control for the method's sensitivity. Strikingly, we found that neurons lacking SorCS2 have increased secretion of PGRN into the medium (Figure 7M), in accordance with the ability of overexpressed SorCS2 to retain PGRN inside the cells (Figure 7E). For comparison, the conditioned medium from *Gm*^{+/-} neurons showed reduced PGRN concentration. When analyzing the intracellular PGRN pool in the *Sorcs2* KO cultures by immunoblotting, we found a substantial decrease in intracellular PGRN (Figures 7N and S12C). In marked contrast, the level of intracellular granulins was significantly increased in the SorCS2-deficient cultures (Figures 7O and S12C).

To conclude, using gain- and loss-of-function models, we demonstrate that SorCS2 binds PGRN to control and fine-tune the concentration of intracellular and secreted PGRN and the production of lysosomal granulins. Moreover, we showed that SorCS2 localization at the plasma membrane is required for PGRN to induce its neurotrophic functions. We

integrate our findings into a proposed model (Figures S13A and S13B).

DISCUSSION

We discovered that SorCS2 receptor is expressed by newborn MNs, where it controls cell-fate decisions (Figure 4), axon outgrowth, and muscle targeting (Figures 1, 2, 3, and 5), as well as synaptogenesis at NMJs (Figure 2). We demonstrate that SorCS2 binds PGRN to regulate its secretion and signaling to enable MN development (Figure 7, model in Figure S13). Our data suggest that these events are controlled by PGRN that is secreted by the MNs themselves and by primitive macrophages juxtaposed to the projecting SorCS2-positive motor axon (Figure 5). These findings are in accordance with previous studies that have linked PGRN to proliferation, neurogenesis, neuronal maturation, axon outgrowth and branching, and synaptogenesis.^{23,24,26,74–86} We further propose that during motor nerve regeneration, the same molecular mechanisms are likely reactivated, as *Sorcs2* KO mice fully phenocopy *Gm* KO mice³⁰ (Figure 6).

SorCS2 binds PGRN intracellularly to control its trafficking and secretion (*cis* interaction)

Reduced activity of lysosomal and extracellular PGRN compromises neuronal development and survival^{87–89} while increasing neuroinflammation.^{25,90} On the flip side, enhanced PGRN promotes cancer growth and metastasis, and thus the risk of lethal outcome.^{38–40} Therefore, the concentrations of extracellular and intracellular PGRN must be tightly regulated. SorCS2 and PGRN are co-expressed by pre- and postnatal MNs (Figures 1, 3, 5, and 6; supplemental figures), where SorCS2 binds PGRN to control its sorting and secretion (*cis* interaction) (Figure 7). We previously reported that PGRN is internalized and trafficked to lysosomes by sortilin, a paralog of SorCS2.⁴³ Notably, deficiency for either receptor increases PGRN extracellularly while depleting the neuronal pool. Despite the structural similarity of VPS10p-D receptors, here we show that this is accounted for by two distinct mechanisms. While *sortilin* KO neurons display reduced PGRN uptake, *Sorcs2* KO neurons exhibit increased PGRN release, since SorCS2 retains intracellular PGRN in TGN. Hence, the increased accumulation of PGRN in the spinal cord of *Sorcs2* KO mice is likely extracellular.

Figure 7. SorCS2 binds PGRN to control its trafficking and secretion

(A and B) IF of HEK293 cells overexpressing hPGRN and hSorCS2 as co-culture (*trans* configuration) (A) or co-expressing the plasmids in the same cell (*cis* configuration) (B). hPGRN and hSorCS2 do not co-localize in *trans* (A), but they do in vesicular structures and plasma membrane when in *cis* (arrows) (B). Hoechst-labeled nuclei (blue).
(C) Co-IP of transfected HEK293 cells. SorCS2 binds PGRN in *cis* but not in *trans*. Arrows point at SorCS2 isoforms with the proform binding hPGRN. β -Actin served as a loading control for the input and negative control for the co-IP.
(D) Immunoblotting of secreted hPGRN in transfected HEK293 cells.
(E–G) Compiled quantification of independent input samples from lysates and cell media blotted for hPGRN and HA-hPGRN. $n = 4/5$. When in *cis*, SorCS2 retains hPGRN intracellularly (E) while reducing PGRN secretion (F). (G) Total PGRN levels are decreased in *cis* but not in *trans* configuration.
(H and I) (H) Immunoblotting of spinal cord from P3 pups ($n = 4$ pups/genotype). (I) The band intensity is normalized to the loading control.
(J–L) Confocal scanning and analysis of PGRN⁺ fluorescent signal in posterior coronal sections of brachial MNs at E12.5. $n = 6$ embryos/genotype. (J) IF staining of PGRN (left panels) was segmented in Imaris (right panels) as spots (gray). (K and L) LMC and MMC populations were either pooled ("all MNs") or distinguished by *Isl1* and *Foxp1* IF. (K) Quantification of PGRN⁺ particle count. (L) Quantification of total volume of PGRN⁺ particles.
(M) ELISA of conditioned media from primary cortical neuron cultures (DIV 7). *Gm*^{+/-} served as semi-negative control. $n = 3$.
(N and O) Immunoblot quantifications of lysates from the cortical cultures ($n = 4$).
All graphs shown mean $\pm$ SEM. See also Figures S11 and S12; Video S10.

SorCS2 is a PGRN receptor enabling its neurotrophic signal transduction (*trans* interaction)

Cultured *Sorcs2* KO MNs failed to respond to PGRN by increased neurite outgrowth. This suggests that PGRN binding to SorCS2 is indispensable for the PGRN neurotrophic effect. Surprisingly, we failed to demonstrate *trans* interaction in SorCS2-transfected HEK293 cells. Conceptually, VPS10p-D receptors require co-receptors to produce high-affinity ligand binding sites to promote processes such as growth-cone collapse or apoptosis.^{10–12,17,91} For instance, the affinity of proNGF to sortilin is increased 20-fold when in complex with p75^{NTR}.¹¹ Similarly, binding of proBDNF to SorCS2 or p75^{NTR} individually is barely detectable, but when the receptors are co-expressed the surface labeling is greatly enhanced.¹² Such a co-receptor may be absent in HEK293 cells but present in cultured MNs that responded to PGRN. The identity of such a co-receptor remains to be determined but, given that PGRN binds Notch receptors to promote MN survival and nerve regeneration,^{92–94} it makes them good candidates.

SorCS2 selectively controls cell-fate decisions and axon outgrowth of spinal cord MNs

SorCS2 regulates axonal innervation via its binding to proneurotrophins proBDNF or proNGF by which it controls growth-cone collapse of CNS neurons.^{12,17,18} Consequently, *Sorcs2* KO mice exhibit dopaminergic hyperinnervation of the prefrontal cortex and attention deficit hyperactivity disorder-like behavior.¹² The repurposing of SorCS2 in this study suggests that SorCS2 may sustain either outgrowth or retraction in a cell-type-specific manner. Alternatively, it may control two sequential processes that jointly enable neurons to first grow their axon and subsequently retract the growth cone for precise targeting. Although we demonstrate that PGRN has direct effect on MN outgrowth that is reliant on SorCS2 expression (Figures 5K–5M) and that PGRN is not expressed in the myotome in either zebrafish or mice (Figures 5A–5H), we cannot formally discount that SorCS2 expression in mesoderm (Figures 1C and 3A) may contribute to the phenotype.

SorCS2 determines cell-fate decisions of MNs. This likely contributes to the impaired innervation of corresponding posterior brachial nerves. The dependence on cell-surface receptor expression and distribution along the body axis during MN diversification and axon outgrowth has been previously reported.^{2,8} It is suggestive that the PGRN-SorCS2 signaling axis controls LMCI neurogenesis, since PGRN also promotes proliferation of retinal and hippocampal neurons during development and in adulthood after injury.^{77,78,82–85} Notably, LMCI neurons did not exhibit high levels of SorCS2 and displayed rather consistent immunoreactivity for intracellular PGRN. Our data thus indicate that SorCS2 has diverse, concentration-dependent functions within different MN clusters. This indicates engagement of additional extrinsic or intrinsic factors interacting with SorCS2 during MN neurogenesis and maturation, which will determine the specificity of muscle targeting.

SorCS2 regulates synaptogenesis at NMJs

We further showed that SorCS2 engages in determination of synapse architecture during NMJ development. This is aligned with SorCS2 forming protein assemblies during synaptic plasticity,

events when synapses undergo rapid morphological changes. Moreover, SorCS2 controls synaptic weakening when in complex with proBDNF and p75^{NTR} at the postsynaptic density, while it mediates synaptic strengthening when engaging the BDNF receptor TrkB.¹⁹ SorCS2 not only mediates synaptic targeting of TrkB but also of NMDA receptors.²¹ Interestingly, PGRN loss of function reduces GluN2B-containing NMDA receptor density and dendritic arborization in murine cortical neurons,⁹⁵ suggesting that PGRN and SorCS2 have other protein partners in common.

SorCS2 deficiency impairs behavior of primitive macrophages associated with the motor axons

We show that PGRN is highly expressed by primitive macrophages from E10.5 that become associated with the motor axons, suggesting that macrophages secrete PGRN toward SorCS2-positive motor nerve to mediate correct pathfinding. Previous studies demonstrated that microglia express and secrete high levels of PGRN in close proximity to neurons,^{96,97} especially when activated upon injury or during MN disease.^{98–102} *Sorcs2* KO mice exhibited an increased number of macrophages associated with distal nerve but, additionally, they displayed a 10% increase in macrophages with low-to-zero PGRN. This suggests that macrophages increased PGRN secretion toward the motor nerve to compensate for the blunted signaling in the SorCS2-deficient MNs. PGRN is also secreted by activated Schwann cells following sciatic nerve injury to promote regrowth of their neighboring MNs.¹⁰³ However, we detected low-to-zero PGRN in Schwann cell precursors, demonstrating that the primitive macrophages/microglia are the primary source of secreted PGRN at embryonic stages.

Perspective on therapeutic opportunities

We have identified SorCS2 as a PGRN receptor required for MN development *in vitro* and *in vivo*. At the plasma membrane, SorCS2 enables PGRN signaling, whereas inside the cell it controls PGRN sorting, secretion, and the production of granulins. SorCS2 can retain non-physiological PGRN levels intracellularly to avoid its excessive release. Since SorCS2 is expressed in several neuronal subtypes, PGRN-SorCS2 signaling likely functions beyond MNs to sustain neuronal integrity. Indeed, we found that SorCS2-deficient cortical neurons that degenerate in patients suffering from frontotemporal dementia caused by *GRN* haploinsufficiency have depleted intracellular PGRN in favor of enhanced extracellular PGRN and intracellular granulins. We therefore propose SorCS2 as a candidate target in the development of disease-modifying treatments for MN diseases, frontotemporal dementia, and other disorders with aberrant PGRN or SorCS2 activity.

Limitations of the study

Our data from human, mouse, and zebrafish models are highly complementary, suggesting a conserved cell-molecular mechanism of PGRN-SorCS2 signaling during neurodevelopment (Figure S13). However, SorCS2 deficiency resulted in prematurely branched motor axons in zebrafish, while branching was not affected in mice. Such a comparison is challenging, since teleost fish differ from higher vertebrates, i.e., in organization of MN clusters. For example, zebrafish MNs do not segregate into an LMC,¹⁰⁴ the population selectively affected in *Sorcs2* KO mice. Since

embryonic tissue adjusts the spatiotemporal programs cell-autonomously based on a complex integration of signals,¹⁰⁵ we cannot exclude the activation of compensatory mechanisms in mice. Furthermore, whether SorCS2 controls neurogenesis in zebrafish and other lower species, and whether its function remains so selective, is to be addressed. Future studies should also thoroughly investigate PGRN-SorCS2 signaling during neurological trauma and regeneration, as it might be relevant for regenerative medicine.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.celrep.2023.113333>.

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AUTHOR CONTRIBUTIONS

Conceptualization: A.N., A.S., and P.B.T.; methodology: A.S., P.B.T., L.W., and J.L.; validation: A.S., K.K.-S., L.W., J.T.-J., S.N., and A.-C.U.; formal analysis: P.B.T., A.S., K.K.-S., L.W., J.T.-J., S.A., M. Kavková, and P.Q.; investigation: P.B.T., A.S., K.K.-S., L.W., J.L., H.L., J.T.-J., S.A., S.B., M.K., P.L.O., S.N., and B.V.; resources: A.N., J.K., P.V.D., J.D., and C.O.; writing – first submission: A.N. and P.B.T.; writing – final manuscript: A.N. and A.S.; visualization: A.S., P.B.T., L.W., and J.L.; supervision: A.N., A.S., K.K.-S., J.K., C.O., H.L., L.N.N., P.V.D., M.V.C., and J.D.; project administration: A.N.; funding acquisition: A.N., J.K., and P.V.D.

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The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Beta-actin	Sigma Aldrich	Cat# A5441; RRID:AB_476744
ChAT	Merck	Cat# AB144P; RRID:AB_2079751
Foxp1, rabbit	Gift from J. Dasen	N/A
Foxp1, guinea pig	Gift from J. Dasen	N/A
GAPDH	Cell Signaling	Cat# cs-2118; RRID:AB_561053
GFP	Abcam	Cat# Ab-290; RRID:AB_303395
HA	Abcam	Cat# Ab-18181; RRID:AB_444303
HA	Sigma Aldrich	Cat# H6908; RRID:AB_260070
HA	Sigma Aldrich	Cat# 26183; RRID:AB_10978021
Hb9, rabbit	Gift from J. Dasen	N/A
hPGRN	R&D Systems	Cat# AF2420; RRID:AB_2114489
hPGRN	R&D Systems	Cat# MAB24201
hSorCS2	DAKO, custom-made	F7100
Iba-1	Wako	Cat# 019-19741; RRID:AB_839504
Islet1/2, concentrate	Developmental studies hybridoma bank	Cat# 39.4D5 (c = 222 μ g/mL); RRID:AB_2314683
Islet1/2, serum	Developmental studies hybridoma bank	Cat# 39.4D5 (c = 30 μ g/mL); RRID:AB_2314683
mPGRN	R&D Systems	Cat# AF2557; RRID:AB_2114504
mSorCS2	R&D Systems	Cat# AF4237; RRID:AB_2192264
Olig2	Abcam	Cat# 109186; RRID:AB_10861310
Pea3	Gift from J. Dasen	N/A
SMI-32 (neurofilament H)	Biolegend	Cat# 801702; RRID:AB_2715852
Sox10	Abcam	Cat# Ab-227680; RRID:AB_2927464
Tuj1	Biosite	Cat# MMS-435P; RRID:AB_2315514
Znp-1	Developmental studies hybridoma bank	Znp-1; RRID:AB_2315626
Chemicals, peptides, and recombinant proteins		
Bungarotoxin (α BTX)	Thermo Fisher Scientific	Cat# B35451; RRID:AB_2617152
Collagenase	Sigma Aldrich	Cat# C9891
Proteinase K	Thermo Fisher Scientific	Cat# EO0491
Target retrieval solution	DAKO	Cat# S169984-2
Tissue-TEK O.C.T.	Sakura	Cat# 4583
NuPAGE LDS sample buffer	Thermo Fisher Scientific	Cat# NP0007
Normal goat serum	Jackson Immune Research	Cat# 005-000-121
Normal donkey serum	Jackson Immune Research	Cat# 017-000-121
Fetal bovine serum	Sigma Aldrich	Cat# F0392
Prolong glass antifade mounting solution	Thermo Fisher Scientific	Cat# P36984
Prolong gold antifade mountant	Thermo Fisher Scientific	Cat# 33342
Fluorescent mounting medium	DAKO	Cat# S3023
Hoechst	Thermo Fisher Scientific	Cat# 33342
cOmplete Protease Inhibitor Cocktail	Merck/Roche	Cat# 11697498001
EDTA	Merck	Cat# 1.08418.0100
Sodium deoxycholate	Merck	Cat# D6750

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Sodium dodecyl sulfate	Merck	Cat# L6026
DTT	Sigma Aldrich	CAS# 27565-41-9
RIPA buffer	Merck	Cat# T9284
Tween 20	Applichem	Cat# A1389.0500
Triton X-100	Applichem	Cat# A1388.0500
Bovine serum albumin	Sigma Aldrich	Cat# 9048-46-8
Sodium azide	Ampliqon	Cat# AMPQ52300
Dextran sulfate sodium salt	Sigma Aldrich	Cat# RES2029D-A7
PhosSTOP (phosphatase inhibitors)	Merck/Roche	Cat# 4906837001
Isoflurane	VWR	CAS# 26675-46-7
BDNF (used in cell cultures)	Merck	Cat# GF029
PGRN (used in cell cultures)	Adipogen	Cat# AG-40A-0188Y
PGRN (used for plasmon resonance)	R&D Systems	Cat# 2420-PG-050
Netrin-1	R&D Systems	Cat# 1109-N1-025
UNC5H4	R&D Systems	Cat# 1429-UN-050
FLRT2	R&D Systems	Cat# 2877-FL-050
FLRT3	R&D Systems	Cat# 2795-FL-050
EphrinA1	R&D Systems	Cat# SMPK3-602-A1
EphrinA2	R&D Systems	Cat# SMPK3-603-A2
EphrinA3	R&D Systems	Cat# SMPK3-359-EA
EphrinA4	R&D Systems	Cat# SMPK3-369-EA
EphrinA5	R&D Systems	Cat# SMPK3-374-EA
EphrinB1	R&D Systems	Cat# SMPK3-473-EB
EphrinB2	R&D Systems	Cat# SMPK3-496-EB
EphrinB3	R&D Systems	Cat# SMPK3-7924-EB
EphA1	R&D Systems	Cat# SMPK1-638-AI
EphA2	R&D Systems	Cat# SMPK1-639-A2
EphA3	R&D Systems	Cat# SMPK1-640-A3
EphA4	R&D Systems	Cat# SMPK1-641-A4
EphA5	R&D Systems	Cat# SMPK1-541-A5
EphA6	R&D Systems	Cat# SMPK1-607-A6
EphA7	R&D Systems	Cat# SMPK1-608-A7
EphA8	R&D Systems	Cat# SMPK1-454-A8
EphB1	R&D Systems	Cat# SMPK2-1596-B1
EphB2	R&D Systems	Cat# SMPK2-467-B2
EphB3	R&D Systems	Cat# SMPK2-432-B3
EphB4	R&D Systems	Cat# SMPK2-446-B4
EphB6	R&D Systems	Cat# SMPK2 -611-B6
Critical commercial assays		
Mouse Progranulin Quantikine ELISA kit	R&D Systems	Cat# MPGRN0
KOD HotStart polymerase	Novagen/Sigma Aldrich	Cat# 71086
GammaBind G Sepharose beads	Sigma Aldrich	Cat# 17061801
Immobilon Forte Western HRP substrate	Sigma Aldrich	Cat# WBLUF0100
ECL Amersham	Sigma Aldrich	Cat# RPN2106
Super SignalTM West Femto maximum sensitivity substrate	Thermo Fisher Scientific	Cat# 34094
DIG RNA labeling kit	Roche	Cat# 11175025910
RNeasy Mini Kit	Qiagen	Cat# 74104
RNA MinElute kit	Qiagen	Cat# 74204

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
mMessage mMachine T7 Transcription kit	Thermo Fisher Scientific	Cat# AM1344
Titanium One-Step RT-PCR kit	Clontech	Cat# 639503
TaqMan Fast Advanced Master Mix	Applied Biosystems	Cat# 4444557
RNA scope Multiplex Fluorescent Detection kit v2	ACD Biotechnie	Cat# 323110
RNA scope Sorcs2-C1 probe	ACD Biotechnie	Cat# 473411
RNA scope Grn-C3 probe	ACD Biotechnie	Cat# 422861-C3
HCR v3.0 RNA-FISH set including two individual grna-B3 and two individual sorcs2-B2 probes used in combo reaction, and HCR amplifiers B3 and B2	Molecular Instruments	HCR™ v3.0 RNA-FISH

Deposited data

Single-cell RNA sequencing of zebrafish embryos	Farnsworth et al. ⁴⁶	http://zebrafish-dev.cells.ucsc.edu
Single-cell RNA sequencing of mouse embryonic spinal cord	Delile et al. ⁶⁷	https://github.com/julielile/MouseSpinalCordAtlas
Laser-captured, multi-cell RNA sequencing of postnatal motor neurons (ArrayExpress; experiments/E-MTAB-7320/)	Nichterwitz et al. ⁷³	GEO https://www.ncbi.nlm.nih.gov/geo/accession_number/GSE115706

Experimental models: Cell lines

Human embryonic kidney 293T cell line	ATCC	Cat# CRL-3216
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Experimental models: Organisms/strains

<i>Tg(mnx1:GFP)^{m12}</i> zebrafish line	Zebrafish International Resource Center	Line #ZL1163
<i>AB strain</i> (WT) zebrafish line	European Zebrafish Resource Center	Line #1175
<i>sorcs2^{sa11642}</i> (<i>sorcs2</i> KO) zebrafish line	Zebrafish Mutation Project	https://www.sanger.ac.uk/collaboration/zebrafish-mutation-project/
C57BL/6j BomTac mouse line (WT)	Taconic	Line #B6-J
<i>Sorcs2</i> $-/-$ mouse line	Glerup et al. ¹²	N/A
<i>B6.Cg-Tg(Hlxb9-GFP)1Tmj/J</i> (= <i>Hb9:GFP</i> +) mouse line	Jackson Laboratories	RRID:IMSR_JAX:005029
<i>Hb9:GFP</i> +/; <i>Sorcs2</i> $-/-$ mouse line	This paper	N/A

Oligonucleotides

<i>sorcs2</i> splice-site inhibiting morpholino Sequence: AAATCTTCTGTCACCTACCTCCACA	GeneTools	N/A
Control morpholino Sequence: CCTCTTACCTCAGTTACAATTTATA	GeneTools	N/A
RT-PCR <i>sorcs2</i> primer set detecting spliced <i>sorcs2</i> in zebrafish: 5'-TTTTCGCGACCTGTACCCAGCTG-3' 5'-TAACGCGCTCCTGAAGCAGAGTC-3'	Eurofins Genomic	N/A
RT-PCR <i>sorcs2</i> primer set suitable for detecting <i>sorcs2</i> during early developmental stages in zebrafish: 5'-TTTTCGCGACCTGTACCCAGCTGTGTT-3' 5'-TAACGCGCTCCTGAAGCAGAGTCCATT-3'	Eurofins Genomic	N/A
RT-PCR <i>rp13a</i> primer set (house-keeping control) for zebrafish: 5'-TCTGGAGGACTGTAAGAGGTATGC-3' 5'-AGACGCACAATCTTGAGAGCAG-3'	Eurofins Genomic	N/A
Primer set used to generate <i>sorcs2</i> probe for Alkaline phosphatase WISH for zebrafish: 5'-GAAGACGCTCATCATGGCAG-3' and 5'- TAATACGACTCACTATAG GGAATGAGGATGACGCTGCTGT-3' (T7 promotor sequence is in bold)	Sigma Aldrich	N/A

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
TaqMan probe detecting <i>Sorcs2</i>	Thermo Fisher Scientific	Cat# Mm00473050_m1
TaqMan probe detecting <i>Gm</i>	Thermo Fisher Scientific	Cat# Mm00433848_m1
TaqMan probe detecting <i>Gapdh</i>	Thermo Fisher Scientific	Cat# Mm99999915_g1
TaqMan probe detecting <i>Hprt</i>	Thermo Fisher Scientific	Cat# Mm00446968_m1
Recombinant DNA		
pcDNA3.1-human_SorCS2_Full length	Glerup et al. ¹²	N/A
pCMV3-human_Progranulin	Sino Biological	Cat# HG10826-UT
pCMV3-HA-human_Progranulin	Sino Biological	Cat# HG10826-NY
pcDNA3.1 empty backbone	Gift from Ernest Arenas	N/A
Software and algorithms		
GraphPad Prism software (versions 7, 9, 10)	Dotmatics	https://www.graphpad.com/
Imaris software package, version 10.0.0.	Oxford Instruments	https://imaris.oxinst.com/products/imaris-for-cell-biologists?gad=1&gclid=EAlalQobChMIdO98dGTgQMVJT8GAB21pw1AEAAAYASAAEgKVofD_BwE
Zen Blue software	ZEISS	https://www.zeiss.com/microscopy/en/products/software/zeiss-zen.html
Adobe Illustrator 2023	Adobe	https://www.adobe.com/
Adobe Photoshop 2023	Adobe	https://www.adobe.com/
BioRender platform	BioRender	https://www.biorender.com/
ImageJ software	National Institutes of Health and the Laboratory for Optical and Computational Instrumentation (Schneider et al.) ¹⁰⁶	https://imagej.nih.gov/ij/
Hokawo software	Hamamatsu	U9304
Other		
CUBIC 1	Susaki et al. ¹⁰⁷	N/A
CUBIC 2	Susaki et al. ¹⁰⁷	N/A
NuPAGE 4–12% Bis-Tris Gels	Invitrogen	Cat# NP0321box
Cryostar NX70	Axlab/Thermo Scientific	Cryostat NX70
Fujifilm Las4000 imaging apparatus	GE Healthcare	LAS 4000
Laser-scanning confocal microscope LMS780 and 800	ZEISS	LSM780/800
Light-sheet 7 microscope	ZEISS	ZEISS Lightsheet 7
Microm HM 560 cryostat	Thermo Fisher Scientific	Microm HM 560 cryostat
Apotome 2.0 microscope	ZEISS	Apotome 2.0
VersaMaxTM tunable microplate reader	Molecular Devices	VersaMaxTM
Olympus IX71 microscope	Olympus	Olympus IX71
Axio zoom V16 microscope	ZEISS	Axio zoom V16
Nikon Ti Eclipse automated TIRF microscope	Nikon	Ti Eclipse
7500 Fast Real-Time PCR system	Thermo Fisher Scientific	N/A

RESOURCE AVAILABILITY

Lead contact

Requests for further information, resources and reagents should be directed to the lead contact, Anders Nykjaer (an@biomed.au.dk).

Materials availability

Unique reagents generated in this study are available upon request from the [lead contact](#), but may be restricted or require a completed material transfer agreement.

Data and code availability

- All raw data are available upon request from the corresponding authors Alena Salasova (alena.salasova@biomed.au.dk) and Anders Nykjaer (an@biomed.au.dk).
- We generated no original code.
- Any additional information required to repeat the methodology or reanalyze the data reported in this work is available upon request from the corresponding authors Alena Salasova (alena.salasova@biomed.au.dk) and Anders Nykjaer (an@biomed.au.dk).

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Primary motor neuron cell culture

Panning plates were coated with monoclonal anti-murine p75^{NTR} antibody (1.46 µg/mL; #Ab8877 Abcam) ON at 4°C diluted in sterile 10 mM Tris-HCl pH 9.5. Primary motor neurons were prepared from E13.5 mice embryos according to⁶⁹ with minor modifications. Health status of embryos corresponded to that reported for the mouse husbandry. We did not exclude any embryos neither we selected any gender since it is not possible to determine sex at embryonic stages. This secured unbiased analysis, for instance independent of the gender. Isolated motor neurons were counted with Neubauer counting chambers and seeded on Poly-DL-ornithine hydrobromide- (PORN; Sigma Aldrich) and Laminin-coated (#23017-015, Invitrogen) coverslips in Neurobasal medium (Gibco) supplemented with GlutaMax I (Gibco), penicillin/streptomycin (1:100, Gibco), B27 (2%, Invitrogen), 2% horse serum. Cells were cultured at 5% CO₂ and 37°C. On day 1 and 3, 80% medium was changed to either new medium without trophic factors or medium containing 5 ng/mL BDNF (#GF029, Merck Millipore) or 100 ng/mL human PGRN (#AG-40A-0188Y, Adipogen). On day 4, the cultures were fixated with 4% PFA for 20 min at RT, washed and stored in PBS-pH7.4 at 4°C until immunostaining.

Primary cortical neuron cell culture

The cultures were obtained from cortex of *WT*, *Sorcs2*^{−/−}, and *Grn*^{+/−} mouse pups at postnatal day 0 (P0).¹² Health status of pups corresponded to that reported for the mouse husbandry. We did not exclude any pups neither we selected any gender since it is not possible to determine sex at this stage. This secured unbiased analysis, for instance independent of the gender. Pups were euthanized by decapitation, brains removed, and cortex dissected into ice-cold Leibowitz's L-15 medium (Life Technologies, #11415049). Tissue was dissociated for 30–40 min using 20 U/mL pre-activated papain (Bionordica, #WBT-LS003126), and then washed in DMEM containing 0.01 mg/mL DNase I (Merck Millipore, #DN25) and 10% FBS before being triturated in the same buffer. Following trituration, DMEM was substituted with Neurobasal-A Medium (Gibco, #10888022) supplemented with 2% B-27 (Gibco, #17504044), 2 mM GlutaMAX (Gibco, #35050), 100 µg/mL primocin (Invivogen, ant-pm-2), 20 µM flouxuridine, and 20 µM uridine (Merck Millipore, #F0503 and #U3750). Cortical neurons were seeded onto pre-coated wells with poly-L-lysine (Merck Millipore, #15140) and laminin (Invivogen, #23017-015). Besides the coating and the culture medium, the cells were grown in naive conditions prior the sample collections. Neurons were incubated at 37°C for 7 days prior to assaying. Neurons were kept in the same culture medium for the last 4 days before assaying. On culture day seven (DIV7), the culture medium was collected for ELISA.

Immortalized cell line culture

Human embryonic kidney 293T cells (HEK293T; ATCC, #CRL-3216) were grown at 37°C with constant 95% of H₂O and 5% of O₂. Cells were cultivated in Dulbecco's Modified Eagle's medium (DMEM; Sigma Aldrich) supplemented with 10% FBS and 50 U/mL penicillin/streptomycin (all Thermo Fisher Scientific). The cells were routinely tested for mycoplasma-free status using MycoAlert Plus Mycoplasma Detection kit (Lonza). The cells were grown in naive conditions. The sex of the line was not determined by ATCC but the gender does not affect the outcome of our overexpression studies.

Zebrafish husbandry

Zebrafish were maintained according to national guidelines under breeding permits 2017-15-0202-00098 and 2022-15-0202-00169 approved by the Danish Animal Experiments Inspectorate (Ministry of Food, Agriculture and Fisheries). Zebrafish were kept under standard conditions with feeding four times a day on a 14 h light/10 h dark cycle at 28.5°C. Zebrafish embryos were not health-tested but were generated by natural crosses of adults from a colony routinely health-monitored using the Zebrafish Comprehensive PCR Panel (Idexx GmbH) with no identified pathogens. The AB strain obtained from the European Zebrafish Resource Center was used to produce *WT* embryos while transgenic line *Tg(mnx1:GFP)^{ml2}* from Zebrafish International Resource Center was used to visualize primary motor neurons. We used heterozygous embryos of *Tg(mnx1:GFP)^{ml2}* line (also referred to as *mnx1:GFP* line in the main text) to avoid any possible cytotoxicity of high GFP expression. Zebrafish line with the mutant allele *sorcs2*^{sa11642} carrying a nonsense mutation within exon 10 in *sorcs2* gene was obtained from the Zebrafish Mutation Project.¹⁰⁸ This line served as a *sorcs2* KO line. All genotypes were confirmed by genomic DNA sequencing.

Genotyping of mutant fish was performed by isolating genomic DNA from adult fish by swabbing¹⁰⁹ and by running PCR using KOD HotStart polymerase (Novagen) following standard protocol with 100 ng of DNA, 5% DMSO, and primers 5'-TATGCACACCCTGATCTT

ATGG-3' and 5'-CACACATACCTGGCTGTTGTTT-3'. Sequencing of PCR products at Eurofins Genomics was used to detect the specific point mutation in the mutant line. Embryos were collected after mating, staged by hpf,¹¹⁰ and kept in E3 buffer (5 mM NaCl, 0.17 mM KCl, 0.33 mM CaCl₂, 0.33 mM MgSO₄, 0.001% methylene blue, 2 mM HEPES, pH 7.2) at 28°C corresponding to naive conditions. In this study, we examined embryos at 6, 12, 24, 27, 36, and 48 hpf. We did not exclude any embryos during any experimental procedure. Neither had we selected any gender since it is not possible to determine sex at these embryonic stages. This secured unbiased analysis, for instance independent of the gender.

Mouse lines genetics and husbandry

All breeding and experiments were approved by the Danish Animal Experiments Inspectorate under permits 2016-15-0202-00051, 2017-15-0201-01192 and 2021-15-0201-00850, and carried out according to institutional, national and European guidelines (FELASA).¹¹¹ The following mouse lines were used in this study: C57BL/6j BomTac *WT* mice (Taconic), *Sorcs2*^{−/−} (*Sorcs2* KO),¹² and heterozygous B6.Cg-Tg(Hlxb9-GFP)1Tmj/J line (referred to as *Hb9:GFP*+/+ in the main text) (Jackson Laboratories). We derived *Hb9:GFP*+/+; *Sorcs2* KO mouse line by crossing the two independent lines. Animals were bred and group-housed (except for adult males used for time-mating) in the classified areas of Animal Facility at Biomedicine, Aarhus University, DK. To ensure good health status prior experimentation and sample collection, the air in the sealed animal facility has been filtered to prevent the occurrence and spread of diseases between the individual units. General health check of mice has been performed and reported twice a year in accordance with FELASA recommendations.¹¹² This included serological, bacteriological, and parasitological diagnostics of animals and the environment. Animals were housed with light-dark cycles of 12 h intervals. Mice were fed standard chow and water *ad libitum*. The housing cages were cleaned every week and supplied with nesting, bedding material, and a tunnel. Genomic PCR confirmed genotypes of all used animals.

We collected samples from these stages: E9.5 – E10.5 – E11.5 – E12.5 – E13.5 – P0 – P3 – 12 weeks old (adult). Embryos were collected from time-mated (6 p.m. – 6 a.m.), plug-positive breeding females. In case of embryos and early postnatal stages, we did not exclude any samples neither we selected any gender since it is not possible to determine sex at these stages. We also did not exclude any adult samples neither selected for gender prior biochemical analyses. This secured unbiased evaluation, for instance independent of the gender. Regarding the facial nerve crush injury, we performed the experiment on both genders with the same result, and presented just males in this article due to space limitations. All samples represented naive (drug-free) tissue unless used for primary MN cultures (see above).

METHOD DETAILS

Experimental design

Experimental design for each method and analysis is described in detail in the main text and below. We have strictly adhered to correct research ethics, performing all experiments with biological and technical replicates, and suitable positive and negative controls. We did not exclude any samples unless identified as statistical outliers. Samples used to assess embryogenesis always included embryos from different breeding mothers since the exact time of oocyte fertilization during time mating varies in both zebrafish and mice. Several experiments and biological replicates were performed blinded and by different researchers and/or in different laboratories using various tools. Such strategy secured solid reproducibility of the data and unbiased results. For instance, quantifications of zebrafish embryos were performed as following: K.K.-S. optimized microinjections; P.B.T. performed the first experiments observing the shorter and more branched axons, and scored the growth cone (Figure S1). For the main data (Figures 1 and 2), K.K.-S. performed multiple new microinjections using WT and transgenic lines, prepared fresh resources, evaluated staging, and fixated the embryos. L.W. took over the samples and performed blinded IF and confocal imaging. A.S. established the semi-automatized Imaris-based segmentation used by L.W. for measuring the axon outgrowth. Besides the required replicates necessary to perform any experimental analysis ($n \geq 3$), we did not apply sample-size estimation since our data were not hypothesis-driven and did not require computation. We also did not randomize nor stratify the data since that is not a suitable strategy for our small datasets.

Primary motor neuron immunofluorescence

For immunostaining of the primary motor neurons, coverslips with neurons were permeabilized in PBS +0.1% Triton X-100 (Appl-chem, #A1388.0500), blocked with PBS +10% fetal bovine serum (Sigma Aldrich, #F0392) for 30 min at RT, and incubated with primary antibodies ChAT or Tuj1 (Table S1) diluted in blocking solution ON at 4°C. After washing, coverslips were incubated with secondary antibodies 2h at RT and nuclei stained with Hoechst (0.2 µg/mL; Thermo Fisher Scientific, #33342). Coverslips were mounted with DAKO mounting medium (DAKO, #S3023) and imaged using an Apotome 2.0 microscope (ZEISS).

Primary cortical neuron cultures

PGRN ELISA

Culture medium samples were analyzed using the Mouse Progranulin Quantikine ELISA kit (R&D Systems, #MPGRN0) following the manufacturer's instructions. At the end of the assay, sample optical density (O.D.) was determined on a VersaMax™ tunable microplate reader (Molecular Devices) and a log/log standard curve was generated using VersaMax™ software Softmax pro 5.4 to determine the PGRN sample concentration following the ELISA kit's instructions.

PGRN/GRN immunoblotting

Primary neurons were lysed in Radioimmunoprecipitation assay buffer (RIPA: 50 mM Tris pH 7.4, 150 mM NaCl, 1% Triton X-100 (Merck, #T9284)), and adding 2 mM EDTA (Merck, #1.08418.0100), 0.5% Sodium-deoxycholate (Merck, D6750), 0.1% sodium dodecyl sulfate (SDS) (Merck, #L6026) supplemented with 1x cOmplete Protease Inhibitor Cocktail (Merck, #11697498001). Homogenates were centrifuged at top speed for 5 min at 4°C. Samples diluted to equal concentration were prepared with NuPAGE LDS sample buffer and 0.02 M dithiothreitol (DTT; CAS#27565-41-9), heated for 10 min at 95°C and loaded into NuPAGE 4–12% Bis-Tris gels (Thermo Fisher, # NP0321box). Separation was carried out using MES running buffer (Thermo Fisher, #NP0002). Protein transfer was performed to PVDF membranes using iBlot Transfer Stack (Thermo Fisher, #IB301032 and #IB301031) according to manufacturer's protocol. Membranes were blocked for 1 h at room temperature with blocking buffer (0.01 M Tris-Base, 0.15 M NaCl, 0.05% Tween 20, 5% skimmed milk powder and 0.002% sodium azide) (Ampliqon, #AMPQ52300), and subsequently incubated with primary antibodies diluted in the same buffer overnight at 4°C. HRP-conjugated secondary antibodies (DAKO) were incubated for 1 h at room temperature and membrane development was performed using Enhanced Chemiluminescence (ECL) Western Blotting substrate (GE healthcare, RPN2106) or ECL Prime Western Blotting substrate (GE healthcare, #RPN2236). In order to detect granulins in neuronal samples, the bottom part of the blotted PVDF membrane was cut below the 14 kDa marker, incubated in 4% PFA for 30 min, followed by 5 min boiling in PBS. Blocking, primary and secondary antibody incubations were carried out as described above. Membrane development was performed using Super Signal™ West Femto maximum sensitivity substrate (Thermo Fisher, #34094). Chemiluminescent signal was imaged with Fujifilm Las-4000 imaging apparatus and analyzed using Fujifilm MultiGauge software.

HEK293 transfection and immunofluorescence

Transfection

HEK293 cells were trypsinized in 1 mL of Trypsin_0.05%-EDTA (Thermo Fisher Scientific) and seeded into 10 cm dishes in 40% confluence 24 h prior the experiment. The next day, the cells were transfected using FuGENE 6 transfection reagent (Promega). Transfection mixture was prepared in serum-reduced OptiMEM medium (Thermo Fisher Scientific) using 6 µg of plasmid DNA per dish (3 µg DNA1: 3 µg DNA2). The following plasmid DNAs were used in this study: pcDNA3.1, pcDNA3.1-human_SorCS2_Full length, pCMV3-human_Progranulin (Sino Biological, #HG10826-UT) and pCMV3-HA-human_Progranulin (Sino Biological, #HG10826-NY). Transfection mixture was added to 7 mL of complete cell media and incubated for 24 h. 24 h post transfection, cells were trypsinized in 1 mL of trypsin and seeded into either 10 cm dishes for co-IP experiments (8 mL of fresh media plus 400 µL of cell suspension, or 400 µL + 400 µL of cell suspension for the co-cultures), or into 24-well plates that contained glass cover-slips coated with Poly-L-Lysine (Sigma Aldrich, #P1524). For the coverslips, 3.5 mL of fresh media with 100 µL cell suspension was added for single-transfections, or 50 µL + 50 µL for the co cultures. 500 µL of this mixture was seeded per well. The cells were subsequently grown 24h and harvested 48h post transfection.

Immunofluorescence

Antibodies used in this study are listed in [Table S1](#). Transfected HEK293 cells were harvested for immunofluorescence 48 h post transfection. They were fixated with 4% paraformaldehyde at RT for 15 min. Immunofluorescence was performed accordingly to ¹¹³ as following: The fixed cells were first washed 3 × 5 min in PBS-pH7.4 at RT, and then blocked in PBTA buffer (sterile PBS-pH7.4 with 0.1% BSA and 0.1% Triton X-100) with freshly added 5% normal donkey serum at RT for 45 min. The primary antibodies were freshly diluted in PBTA-5% donkey serum according to [Table S1](#), and incubated at +4°C overnight. The cells were then washed 3 × 5 min in PBS-pH7.4 at RT. Afterward, the samples were incubated at RT for 1 h with Alexa-Fluor conjugated secondary antibodies (Thermo Fisher Scientific) diluted 1:800 in PBTA-5% donkey serum together with Hoechst (1:2000; Thermo Fisher Scientific, #33342). The samples were then washed 3 × 5 min in PBS-pH7.4 and mounted with DAKO fluorescent mounting solution (DAKO, #S3023). The samples were imaged using laser-scanning confocal microscopy system ZEISS LSM-780.

Co-immunoprecipitation and immunoblotting in HEK293 cells

Antibodies used in this study are listed in [Table S1](#). For the co-IP experiments, HEK293 cells were harvested either 24 or 48h post transfection. First, 2 mL of cell media was spun down at 200g for 5 min at RT to remove the cell debris. The samples were placed on ice, and cells were washed with PBS before adding 1 mL of TNE lysis buffer (10mM Tris, 150 mM NaCl, 1mM EDTA, 1% NP40, pH = 8) with freshly added phosphatase (Merck, #4906837001) and protein inhibitors (Merck, #11697498001). The cells were lysed while shaking at 4°C for 20 min. Cell lysates and cell medium were spun down at 20,000 g at 4°C for 15 min, and supernatant was kept for further processing. 60 µL from each sample was used as an INPUT reference sample and was mixed with 20 mM DTT and NuPAGE LDS sample buffer. 400–600 µL of the cell lysate per condition were used for the pulldown. The following antibodies were used: 1 µg of progranulin (R&D systems, #AF2420), 1.5 µg of rabbit anti-HA (Sigma Aldrich, #H6908), and 1 µg of mouse-anti-HA (Sigma Aldrich, #26183). Cell lysates were incubated with antibodies for 14–16 h in 4°C. Meanwhile, 30 µL GammaBind G Sepharose beads (Sigma Aldrich, #17061801) per sample were washed with PBS, centrifuged at 300g for 5 min at RT, and washed in TNE lysis buffer before another centrifugation step at 300g for 5 min at RT. Washed beads were incubated with the antibody-lysate for 4 h at 4°C with slow rotation. Samples were subsequently washed on ice with PBS +0.1% Tween 20 (Applichem, #A1389.0500) and spun down at 200g for 1 min at 4°C. Proteins were eluted by heat in elution buffer (TNE buffer, 20 mM DTT, NuPAGE LDS sample buffer). All the samples (including INPUT) were immediately boiled at 95°C for 5 min and subjected to immunoblotting using

home-made 8% Tris-Glycin polyacrylamide SDS-Page gels. Immunoblotting was performed as described in the method section “P3 and adult spinal cord experiments – immunoblotting”. Secondary antibodies fused with HRP (DAKO) were used in dilution 1:2000. The immunoblots were developed dependent on the strength of the signal either with ECL Amersham (Sigma Aldrich, #RPN2106) or Immobilon Forte Western HRP substrate (Sigma Aldrich, #WBLUF0100). Images were processed using Adobe Photoshop CC. The band intensity was measured as peak area (square pixel = 1) in ImageJ. The band intensities of IB: PGRN and IB: HA were further normalized to the band intensity of β -actin (loading control). To describe relative band intensity, the values were further normalized to IB of Progranulin single transfection, which was set as 1.

Zebrafish microinjections

The *sorcs2* splice site-inhibiting MO (sMO) targeted the exon 2/intron 2 boundary (AAATCTTCTGTCACTTACCTCCACA) and standard control MO (cMO) were obtained from GeneTools, LLC. Given an injection volume of 5 nL, injection mixes were made with phenol red (0.2% in 1 mM HEPES, pH7.5), 5 ng of p53 MO to avoid unspecific apoptotic effects,¹¹⁴ and either 5 ng of cMO or 3.7 ng of the *sorcs2* sMO. For rescue experiment, we injected 500 pg of human *SORCS2* mRNA was injected together with sMO. Injections were performed under a microscope using a pico-injector in embryos from the 1 to 2-cell stage. Human *SORCS2* mRNA was transcribed from a pcDNA3.1-human_SorCS2_Full length plasmid (provided by Peder S. Madsen, Aarhus University) with mMessage mMachine T7 Transcription kit (Thermo Fisher, #AM1344) and purified with RNA MinElute kit (Qiagen).

RT-PCR of zebrafish larvae

RNA was purified from 30 to 50 embryos using RNeasy Mini Kit (Qiagen) and RT-PCR was performed to detect *sorcs2* mRNA in WT embryos using the Titanium One-Step RT-PCR kit (Clontech) with primers 5'-TTTTCGCGCACCTGTACCCAGCTG-3' and 5'-TAACGCGC TCCTGAAGCAGAGTC-3' for detection of mRNA after injection of different amounts of sMO. We used 5'-TTTTCGCGCACCTGTACC CAGCTGTGTT-3' and 5'-TAACGCGCTCCTGAAGCAGAGTCCATT-3' to detect *sorcs2* mRNA in the different developmental stages. As a reference, the *rpl13a* gene was used with primers 5'-TCTGGAGGACTGTAAGAGGTATGC-3' and 5'-AGACGCACAATCTTGAGAG CAG-3'. All primers were generated at Eurofins Genomic.

Whole-mount RNA in situ hybridization of zebrafish larvae

Alkaline phosphatase WISH

Alkaline phosphatase WISH was carried out according to the Nature Protocol by Thisse and Thisse¹¹⁵ with addition of 5% Dextran sulfate sodium salt (Sigma Aldrich) to the hybridization mixture.¹¹⁶ The *sorcs2* probe was transcribed using the DIG RNA Labeling Kit (Roche, #11175025910) with a T7 polymerase from a PCR fragment made with the following primers on embryonic cDNA (the T7 promoter sequence is shown in bold): 5'-GAAGACGCTCATCATGGCAG-3' and 5'-**TAATACGACTCACTATAGG**GAATGAGGAT GACGCTGCTGT-3' (Sigma Aldrich). Stained embryos were stored in 75% glycerol and images acquired using an Olympus IX71 microscope with an Olympus DP71 camera. Horizontal sections were obtained by molding stained embryos in Tissue-TEK O.C.T. (Sakura) on dry ice and sectioning on a Microm HM 560 cryostat.

Fluorescent WISH

Whole-mount HCR v3.0⁴⁹ (Molecular Instruments) was performed with probes and reagents provided by the manufacturer following their protocol, and with HCR-B2 and HCR-B3 amplifiers (Molecular Instruments). Zebrafish embryos were collected at 24, 27, and 48 hpf and fixated with 4% PFA ON at 4°C. 48 hpf embryos were furthermore bleached 1 h with 3% H₂O₂/0.5% KOH/PBS before dehydration to remove pigmentation. Following rehydration, the 48 hpf embryos were then treated with Proteinase K (Thermo Fisher Scientific) for 20 min at RT and post-fixated 20 min RT. All embryos were incubated with 5 pmol/ μ L of probes ON at 37°C and hairpins as instructed. Two set of probes for *sorcs2*-B2 and *grna*-B3 were used in combo to increase the signal. After washing and removing the yolk sac,¹¹⁷ embryos were mounted laterally with Prolong Glass Antifade Mountant (Thermo Fisher Scientific, #P36984). 10 μ m cross sections were obtained of zebrafish trunk by embedding stained embryos in O.C.T. (Sakura) on dry ice and sectioning on a Cryostar NX70 (Axlabs). Embryos were imaged on a confocal laser-scanning microscope LSM-780 (ZEISS). A 20 $\times$ objective was used to obtain z-stacks with step intervals of 1.9 μ m of trunk motor axons and cross sections.

Whole-mount immunofluorescence and confocal imaging of zebrafish

Immunofluorescence

Zebrafish embryos were dechorionated using sharp forceps, and fixated in 4% PFA ON at 4°C. For a single IF of Mnx1-GFP embryos using AChRs marker Bungarotoxin (BTX), the embryos were washed 3 $\times$ 5 min in 1xPBS and incubated in 1 mg/mL collagenase (Sigma Aldrich, #C9891-100MG) diluted in 1xPBS-pH7.4 containing Ca²⁺ ions at RT for 10 min for 27 hpf or 45 min for 48 hpf embryos, respectively. Following 3 $\times$ 5 min wash in PBS pH7.4, the embryos were incubated at RT for 30 min with 10ug/mL α BTX-555 (Thermo Fisher Scientific, #B35451) in potassium phosphate buffer containing 0.5% Triton X-100, 10% normal goat serum (Jackson Immune Research) and 1% DMSO. After 5 $\times$ 5 min washing steps, the embryos were immersed in 75% glycerol. To label NMJs in WT embryos, we used Znp-1 antibody labeling pre-synaptic Synaptotagmin-2 and α BTX-555 labeling post-synaptic AChRs. After BTX-555 staining and subsequent washing steps, the embryos were incubated with 1.25 μ g/mL Znp-1 antibody (Developmental studies hybridoma bank, #Znp-1, deposited by Trevarrow B., University of Oregon) for 16 h for 27 hpf or 72 h for 48 hpf embryos,

respectively. Following 3 × 5 min washing in 1xPBS, the embryos were incubated with goat anti-mouse Alexa 488 (1:1000, Thermo Fisher Scientific) at RT for 3h. The samples were washed again 3 × 5 min in 1xPBS.

Confocal laser-scanning microscopy

All embryos were immersed in 75% glycerol, and after removing the yolk sac¹¹⁷ mounted in DAKO mounting fluorescence medium laterally for confocal laser scanning microscopy using LSM-780 or LSM-800 microscopes (ZEISS). We used 20x or 40x objectives to obtain z-stacks with step intervals of less than 2.3 μm. The motor neurons were counted rostrally ranging from motor axon #5 to #16. For manual scoring of the growth cone in Znp-1 staining, each z stack was split into the two hemi-segments of the axon rows and z-projected using ImageJ software.^{106,118} The axons were scored using definitions described in¹¹⁹ for length and¹²⁰ for branching.

Live-imaging of *mnx1:GFP* zebrafish larvae

cMO- and SMO-injected *Tg(mnx1:GFP)^{ml2}* embryos were dechorionated and anesthetized with 0.05% tricaine in E3 buffer. They were laterally molded at 18 hpf into 0.5% low-melt agarose in an imaging dish with glass bottom and covered with E3 buffer containing tricaine to keep them sedated during imaging. Embryos were imaged from 19 hpf until 39 hpf with a Nikon Ti Eclipse automated TIRF microscope equipped with a 10x differential interference contrast (DIC) air objective, a Perfect Focus 3 system and a Zyla sCMOS5.5 Megapixel camera (Andor) controlled by NIS Elements software from Nikon. The fluorescence illumination system was Cool LED-pE-300 white, and the fluorescence filter set used was standard GFP. Imaging, using wide-field settings, was performed every 15 min for 20 h with 13 z-stacks separated with steps of 6 μm of both DIC light microscopy and fluorescence microscopy.

Touch response assay of 48 hpf zebrafish larvae

Injected embryos were dechorionated at 47 hpf and returned to incubator for least an hour to avoid stress. In a Petri dish with warmed E3, embryos were placed, one at a time, in the center of concentric circles drawn with an increase in diameter of 0.5 cm and touched with an embryo poker¹¹⁷ at the back from behind. Using a ZEISS Axio zoom V16 microscope with a Hamamatsu high-speed camera, embryos were recorded for 10 s collecting 100 images/second. Images were analyzed using the Hokawo software (Hamamatsu) and time spent swimming 1 cm was determined. Embryos that did not swim 1 cm were not included in analysis.

Immunofluorescence of mouse embryos

Sample preparation

Embryos were removed from the maternal uterus into ice-cold 1xPBS-pH7.4, decapitated (head was used for genotyping), and fixed in 4% formaldehyde while rotating at +4°C. PGRN staining required 24 h long fixation. In contrast, SorCS2 staining required 2–4 h fixation. If fixed otherwise, immunolabeling for these two proteins did not work. Embryos were washed three times in ice-cold PBS-pH7.4. Embryos were then gradually immersed in 10%, 20%, and 30% of saccharose solution (in PBS-pH7.4) at 4°C. Dehydrated embryos were embedded in Tissue-TEK O.C.T. (Sakura) and frozen to –80°C. The tissue was cryo-sectioned (Cryostar NX70, Axlab) as 14 μm slices or as 60 μm slices for macrophage analysis. The sections were air-dried and stored at –80°C.

Immunofluorescence

All antibodies used in the study are listed in Table S1. Prior IF, slides were thawed in RT for 5 min, and washed 2 × 5min in 1xPBS-pH7.4 while gently shaking in RT. The sections were blocked with sterile PBTA solution (0.1% BSA, 0.3% Triton X-100, 1x PBS-pH7.4) containing freshly added 5% donkey serum (Jackson Immuno Research) at RT for 1 h. Primary antibodies were diluted according to Table S1 (“IF tissue”) in PBTA-5% donkey serum and incubated with the slides in the wet-chamber at +4°C for 24 h. The next day, slides were washed 3 × 5min in 1x PBS-pH7.4 and incubated with corresponding Alexa-based secondary antibodies (Thermo Fisher Scientific; diluted as 1:800 together with Hoechst 1:2000) at RT for 2 h. The slides were washed 3 × 5min in 1xPBS-pH7.4, and mounted using 24 mm × 60 mm coverslips with DAKO mounting fluorescence medium or 600 μL of Prolong glass antifade mounting solution (Thermo Fisher Scientific) in case of 60 μm sections. The slides were stored at +4°C for 24 h and scanned at laser-scanning confocal microscope LSM780 or LSM800 (ZEISS). Unless stated otherwise, all images were post-processed in ImageJ and Adobe Photoshop.

Tissue clearing and light-sheet microscopy

Sample preparation

E12.5 embryos were removed from the maternal uterus into ice-cold 1xPBS-pH7.4. While we collected forebrain and dorsal midbrain for genotyping, we fixed the remaining specimen in ice-cold 4% formaldehyde (in 1xPBS-pH7.4) while rotating at +4°C for 24 h. Embryos were then washed 3 × 5min in ice-cold PBS-pH7.4, and were kept in dark at +4°C in 1xPBS containing 0.1% sodium azide until further processing. We analyzed nine *Hb9:GFP/+* embryos from 3 independent breedings, and nine *Hb9:GFP/+; Sorcs2–/–* embryos from 2 independent breedings.

Tissue clearing and light-sheet fluorescence microscopy

To improve the tissue clearing process, the abdominal cavity of the embryos was cut using straight blade eye scissors and the inner organs were gently removed using fine, curved tip tweezers. Eviscerated embryos were then cleared using the CUBIC protocol.¹⁰⁷ Embryos were put into 11 mL of CUBIC1 solution (25% urea, 25% N,N,N',N'-tetrakis(2-hydroxypropyl)ethylenediamine, 15% Triton X-100) after the excess PBS was removed by dabbing the embryos on a paper towel. The 15 mL tubes containing the embryos were then put onto a horizontal shaker and incubated at room temperature for 48 h while protected from light. The embryos were subsequently transferred into fresh 15 mL tubes containing 11mL of CUBIC2 solution (50% sucrose, 25% urea, 10% 2,2',2''-nitrotriethanol, 0.1%

Triton X-100) and incubated on a horizontal shaker overnight at room temperature while protected from light. Cleared embryos were then stored in CUBIC2 at 4°C until imaging. Light-sheet fluorescence microscopy was performed on ZEISS Light-sheet 7 microscope (ZEISS) using a 5× objective. Individual samples were glued to a vertical sample holder using a superglue, and then imaged while submerged in a sample chamber filled with CUBIC2. The embryos were imaged in sagittal orientation with only the half of the embryo closer to the detection objective being imaged.

Fluorescent multiplex v2 RNA scope *in situ* hybridization of mouse embryos

Mouse embryos were rinsed in PBS-pH7.4 (Invitrogen), and fixed in 4% formaldehyde (Acros Organics) at 4°C for 24 h. The tissue was gradually immersed in 10%, 20%, and 30% of saccharose/PBS-pH7.4 solution at 4°C until dehydrated. The embryos were washed, embedded in Tissue-TEK O.C.T. (Sakura), and frozen to –80°C. The tissue was cryo-sectioned (Cryostar NX70, Axlab) as 14 µm slices that were air-dried in –20°C for 2 h, and later stored at –80°C. For RNA ISH itself, we implemented manufacturer instructions for fixed-frozen samples and RNAscope Multiplex Fluorescent Reagent Kit v2 kit (ACD Biotechnie, #323100) using the following probes: Mm-Sorcs2-C1 (ACD Biotechnie, #473411-C1) and Mn-Granulin-C3 (ACD Biotechnie, #422861-C3). For the fluorescent labeling, we used Opal-690 reagent pack (Akoya Biosciences, FP1497001KT, 1:750) for the C1 probe, and Opal-570 reagent pack (Akoya Biosciences, #FP1488001KT, 1:2000) for the C3 probe. To secure good Hb9-GFP signal, we subsequently incubated the slides with rabbit anti-GFP antibody (Abcam, #ab290, 1:1000) in PBTA solution (0.3% Triton X-100, 0.1% BSA, 5% donkey serum) at 4°C ON. The next day, we used secondary antibody anti-rabbit Alexa 488 (Invitrogen, #A11008, 1:1000) together with Hoechst (Thermo Fisher Scientific, #33342, 1:2000) for 1 h in RT. IF was skipped for Hb9 negative *WT* samples. We mounted the slides with Prolong Gold Antifade Mountant (Thermo Fisher Scientific, #33342), and imaged them using laser-scanning confocal microscope LSM-780 and -800 (ZEISS).

P3 and adult spinal cord experiments

Immunofluorescence

We performed transcardial perfusion on adult *WT* and *Sorcs2*^{–/–} mice using 1xPBS-pH7.4 followed by 4% formaldehyde. Spinal cords were then dissected out and post-fixed in 4% PFA for 2–6 h. P3 spinal cord were directly removed from *WT* or *Hb9:GFP*⁺ mice and fixated in 4% PFA ON at 4°C. For both ages, the spinal cords were dehydrated in 30% sucrose ON at 4°C before freezing in Tissue-TEK O.C.T. on dry ice. 40 µm sections of adult tissue were cut on a Microm HM 560 cryostat and 10 µm sections of P3 on a Cryostar NX70 (Axlab), and they were stored at –80°C until further use. For IF staining, slides were thawed and antigen-retrieval performed by incubating the sections with preheated Target Retrieval Solution (DAKO, #S169984-2) 20 min at 70°C. Sections were incubated in Tris-buffered saline (TBS) + 5% donkey serum + 1% bovine serum albumin + 0.3% Triton X-100 for 30 min at RT for permeabilization of cells and blocking. Slides were incubated with primary antibodies ON at 4°C. The subsequent day, the sections were washed with TBS + 0.05% Triton X-100 and incubated with secondary antibodies (1:300, Thermo Fisher) 4 h at RT. After incubation with Hoechst (0.2 µg/mL, Thermo Fisher), the slides were mounted using DAKO mounting medium. Images were acquired on a confocal laser-scanning microscope LSM-780 (ZEISS).

RT-qPCR

Cortex, hippocampus, cerebellum, spinal cord and gastrocnemius muscle from *WT* and *Sorcs2*^{–/–} mice were collected in RNeasy lysis buffer (Invitrogen) and left ON at 4°C. The following day the tissue was lysed in RLT buffer and RNA purified using RNeasy (Qiagen). cDNA was synthesized using the High Capacity RNA-to-cDNA kit (Thermo Fisher). RT-qPCR was performed using TaqMan Fast Advanced Master Mix (Applied Biosystems, #4444557), TaqMan expression probes, and a 7500 Fast Real-Time PCR system (Thermo Fisher). *Hprt* and *Gapdh*, were used for normalization measuring fold change by $2^{-\Delta\Delta C_t}$, and three technical and three biological replicates were analyzed. All TaqMan probes were determined to be within 90–110% efficiency. TaqMan probes used (Thermo Fisher): *Sorcs2* (Mm00473050_m1), *Gm* (Mm00433848_m1), *Gapdh* (Mm99999915_g1), *Hprt* (Mm00446968_m1).

Immunoblotting

Spinal cords from *WT* and *Sorcs2*^{–/–} mice were dissected and lysed on ice with TNE lysis buffer: Tris-base 10 mM, NaCl 150 mM, EDTA 1 mM, NP40 1%, cOmplete 1% (Roche), PhosSTOP 1% (Roche) and mechanically with a pellet pestle. Lysed tissue was centrifuged at 13,000 rpm and protein concentration of the supernatant measured using a bicinchoninic acid assay. Protein samples were denatured by boiling with SDS sample buffer and 20 mM DTT. Proteins were then separated by SDS-PAGE using a precast NuPAGE 4–12% Bis-Tris Gel (Invitrogen) and transferred to a PVDF membrane by immunoblotting in blotting buffer (192 mM glycine, 25 mM Tris-Base, pH 8.0). The membrane was subsequently blocked in 0.05 M Tris-base, 0.5 M NaCl, 0.1% Tween 20 (TST buffer) + 2% skimmed milk powder and 0.5% Tween 20 and washed in wash buffer (10 mM HEPES pH 7.8, 140 mM NaCl, 2 mM CaCl₂, 1 mM MgCl₂, 0.2% skimmed milk powder, 0.05% Tween 20). It was incubated with primary antibodies diluted in wash buffer for 2 h at RT, then HRP-conjugated secondary antibodies (DAKO) and visualized with chemiluminescence using ECL Western Blotting Detection Reagents (GE Healthcare) and the imaging system FUJI Film LAS 4000 (GE Healthcare).

Facial nerve crush injury experiments

Surgery

The facial nerve crush surgery was performed at VIB-KU Leuven and were approved by Institutional Animal Care and Ethical Research Advisory Committee of the University of Leuven, Belgium (P132/2012 and P015/2015). The surgeries were performed according to³⁰ on 10-weeks old *WT* and *Sorcs2*^{–/–} mice of both sexes that were shipped to Belgium from Denmark. Mice were placed

on a heating pad and anesthetized using 3% isoflurane (VWR). The incision area around stylomastoid foramen was shaved and disinfected first. We separated the facial nerve from the connective tissue using blunt forceps and identified the injury site proximally to bifurcation of the retroauricular branch. The facial nerve was gently crushed for 20 s using an ultra-fine hemostat with smooth tips. The incision was sutured with 6-0 absorbable sutures (Ethicon). We returned the mice to their home cages with a heating lamp on to prevent hypothermia. When fully recovered, we validated the positive outcome of the surgery by the absence of the whisker movement on the ipsilateral side of the nerve crush injury.

Immunofluorescence

Immunostaining of the facial motor nucleus was performed according to³⁰. In short, deeply anesthetized mice were subjected to transcardial perfusion, first using PBS-pH7.4 to remove all the blood, and then by injecting 4% Paraformaldehyde (PFA) to crosslink the proteins. The whole brain was removed from the skull and post-fixed in 4% PFA at +4°C overnight. The tissue was subjected to dehydration in 30% sucrose (Sigma Aldrich) at +4°C for 48 h, and then it was cryopreserved using Tissue-TEK O.C.T. (Sakura) for microtome sectioning. 25 μ m thick sections were blocked in PBS-pH7.4 containing 3% bovine serum albumin (Sigma Aldrich) and 0.1% Triton X-100 (Appllichem). Subsequently, the samples were incubated with primary antibodies against SMI-32 (1:200, Biolegend) and SorCS2 (1:100, R&D systems) diluted in PBS-pH7.4 with 1% BSA and 0.1% Triton X-100 at +4°C overnight. The next day, the sections were washed and incubated with Alexa-conjugated secondary antibodies (Invitrogen) diluted in PBS with 1% BSA and 0.1% Triton X-100 at RT for 1 h. The sections were mounted with Prolong Gold antifade reagent (Thermo Fisher Scientific, #33342) and stored at +4°C. Images were acquired using a Leica SP8x confocal microscope.

Surface plasmon resonance

Surface plasmon resonance was performed to determine binding affinity of candidate protein to extracellular domain of SorCS2 using a Biacore 3000 instrument. Soluble SorCS2 was immobilized at a density of 0.0862 pmol/mm² on a CM5 sensor chip. hPGRN was dialyzed against 10 mM HEPES (pH 7.4), 150 mM NaCl, 1.5 mM CaCl₂, 1 mM EGTA, and 0.005% surfactant P20 (running buffer), and 25–250 nM of the ligand was injected (5 mL/min) through the flow cell at 20°C. After 600 s, running buffer containing PGRN was substituted with buffer lacking the ligand, allowing for determination of both the association and dissociation constants. The binding response is expressed in relative response units, i.e., the difference in response between the immobilized protein flow cell and a corresponding control flow cell. Kinetic parameters were determined using the BIA evaluation 3.0 software. The recombinant proteins and the results are listed in Table S2.

QUANTIFICATION AND STATISTICAL ANALYSIS

Primary motor neuron outgrowth analysis

Neurons were measured using ImageJ, and neurites longer than twice the length of the soma and projecting directly from the cell body were considered primary. Secondary neurites were defined as neurites protruding from the primary, and tertiary as neurites protruding from the secondary. 31–39 neurons of each genotype and stimulation were analyzed from two independent experiments.

Imaris-based segmentation of *grna* and *sorcs2* transcripts for co-localization in zebrafish

The fluorescence was segmented in Imaris v10.0.0. (Oxford Instruments) using automatized settings. Mnx1:GFP signal was segmented as surface, while *sorcs2* and *grna* were segmented as spherical spots. Co-localization was determined and filtered as distance of spots from Mnx1+ surface ≤ 0 μ m. After this data filtration, close proximity of *grna* expression was determined and filtered as distance of spots from Mnx1+ surface ≤ 5 μ m while ≥ 0.1 μ m.

Imaris-based segmentation and quantification of *mnx1:GFP*+ axons in zebrafish

Mnx1-GFP-positive axon outgrowth reconstruction and analysis was performed in Imaris v10.0.0. (Oxford Instruments). We used filament segmentation (diameter = 3 μ m) with manual settings for the main filament representing the main axon. The filament was navigated according to the fluorescent signal from the point exiting the spinal cord to the tip of the axon. Individual branches were reconstructed by adding additional seeding points. The filament length was extracted from the Imaris-based statistics for the individual filaments and further analyzed in GraphPad Prism.

27 hpf embryos

We quantified neurons from three independent microinjections; ctr MO - 9 embryos, *sorcs2* sMO - 9 embryos, rescue - 10 embryos. We analyzed 7 axons per embryo and counted motor axons rostrally starting at motor axon #7 to motor axon #13. In total, we reconstructed and analyzed 63 axons for ctr MO condition, 63 axons for *sorcs2* sMO, and 70 axons for the rescue condition.

48 hpf embryos

We quantified neurons from three independent microinjections; ctr MO - 11 embryos, *sorcs2* sMO - 12 embryos, rescue - 10 embryos. Similarly to 27 hpf, we analyzed 7 axons in total, ranging from axon #7 to #13, counted rostrally. In total, 77 axons were analyzed for ctr MO, 84 axons for *sorcs2* sMO, and 70 axons for the rescue.

Quantification of synaptic compartments of zebrafish NMJs

The synaptic puncta analysis was performed in ImageJ according to⁵¹ with a few modifications. Briefly, we created a ROI around each of the analyzed NMJ region at horizontal myoseptum. Then, we set a threshold for each picture using an automatic ImageJ setting (“Yen” and “Otsu” methods for BTX and Znp-1, respectively, and “Otsu” for merged channels). After thresholding, the “Analyze Particle” function was used with a minimum puncta size of 0.6 μm^2 . We quantified samples from three independent micro-injections collected at 27 hpf; ctr MO - 10 embryos, *sorcs2* sMO - 11 embryos, the rescue - 11 embryos. We analyzed 9 axons per each embryo (motor axon #6 to #14, counted rostrally). In total, 90 axons were analyzed for ctr MO, 99 axons for *sorcs2* sMO, and 99 axons for the rescue condition.

Cell-fate tracing of mouse brachial MNs

We analyzed six E12.5 embryos from two independent breedings per each genotype. The embryos were first decapitated, and then fixed in 4% formaldehyde for 24 h as described above. We cryo-sectioned (Cryostar NX70, Axlab) 14 μm thick coronal sections of cervical and brachial spinal cord, which were collected sequentially on 10 constitutive slides, and processed for IF staining and confocal scanning as described above. These primary antibodies were used: gpFoxp1, *clslet1/2*, and Pea3 to distinguish LMCm, LMCI, and MMC motor columns, as well as the Pea3+ motor pool. Pea3 immunoreactivity also determined C4-T1 spinal segments as previously reported,^{60,61} which resulted in seven analyzed Pea3+ sections per embryo which were either presented individually or summed as the final cell count per embryo. Cells were counted in ImageJ using “Cell counter” plugin.

Quantification of macrophage association to mouse motor nerve

We analyzed four *Hb9:GFP/+* and four *Hb9:GFP/+; Sorcs2 KO* E12.5 embryos. The samples were prepared as described for the cell-fate tracing. We cryo-sectioned (Cryostar NX70, Axlab) 60 μm thick coronal sections of entire spinal cord, which were sequentially collected on seven constitutive slides. Each slide contained four cervical and brachial sections, three thoracic sections, and two lumbar sections. We analyzed three sections covering brachial spinal cord with comparable histology among the samples. We performed IF and confocal scanning (10 $\times$ objective, 0.5 zoom) as described above using primary antibodies against mPGRN and Iba1 to determine macrophages expressing high levels of PGRN. Unsaturated z stack confocal tile scan of the whole section was taken as 8–11 slices with 4- μm interval. These stacks were post-processed in ImageJ using stacks/Z-project with maximum intensity. Histologically, we determined three regions dependent on the proximity to the Hb9+ somas: 1. proximal area before first branching (appx. 100 μm from the ventral horn), 2. Medial area (horizontal line until notochord), 3. Distal area. We counted only those Iba1+ cells that were directly associated with the GFP signal. We used Cell counter plugin in ImageJ for the manual counting.

Imaris-based mouse *Hb9:GFP/+* nerve annotation and segmentation of light-sheet datasets

Light-sheet images were post-processed in ZEN Blue 3.7 software (ZEISS). The regions of interest (right forelimb, right hindlimb, and right trunk) were exported as separate subsets of the original data, and converted to “.ims” format for further analysis in Imaris v10.0.0. (Oxford Instruments). Brachial nerves were annotated according to,^{8,56} hindlimb nerves were annotated according to.⁵⁷ We used filament segmentation to reconstruct the nerve and its branches, and to measure their length. We used a feature “measurement points” to address the nerve thickness and perimeter, in some cases also the length. The forelimb/brachial nerve reconstruction and measurements are visualized in (Video S9), the analyses of thoracic and lumbar nerves are visualized in (Figures S4H and S4O). The measurement data (including “straightness”) were extracted from Imaris-based statistics.

Imaris-based segmentation of PGRN+ particles in mouse embryos

Sample preparation

We used 14 μm thick coronal sections from E12.5 embryos analyzed for the cell-fate tracing. We selected three posterior sections closely located to those that were significantly affected in their cell fate decisions. IF staining was performed as described above using these primary antibodies: mPGRN, rbFoxp1, and *slslet1/2*. The sections were scanned using confocal microscope LSM-800 (ZEISS) using 40 $\times$ objective with 0.5 zoom as a z stack image with 1- μm interval.

Semi-automatized segmentation using Imaris v10.0.0

The images were converted into “.ims” file type using ImarisFile converter 10.0.0 (Oxford Instruments). The PGRN signal was segmented as “region of interest” corresponding to the MN domain stained by Foxp1 and *Islet1*. We then selected segmentation as “different spot sizes (region growing)”. We measured that a diameter of one PGRN particle is usually around 0.3 μm –1.3 μm , thus we selected 0.6 μm as estimated XY diameter. We then used “Model SPF-elongation along z axis option” with Z diameter equal 3- μm . We subtracted the background and filtered the spots as “quality feature” above 1411. Local contrast was selected for “spot region type”. The volume of the segmentation was determined in region border with an average value 28. All sections were segmented with identical setting described above, only region of interest was determined manually. We deleted those spots that either belonged to microglia (high PGRN expression) or which belonged to non-MN cells. *Islet1* and Foxp1 staining helped to distinguish LMC from MMC neurons.

Facial nerve crush injury quantification

To evaluate regeneration of facial nerve axons, all mice were examined twice daily (9–10 a.m. and 6–7 p.m.) for whisker movement. Whisker movement scores were given according to the following scale: 0 for no detectable movement, 1 for detectable motion of individual whiskers, 2 for significant (but asymmetric) voluntary motion, and 3 for symmetric voluntary motion. The contralateral whiskers were used as a reference for complete functional recovery.³⁰

Analysis of deposited single-cell and multi-cell RNA sequencing data

The single-cell transcriptome atlas of zebrafish embryos⁴⁵ on <http://zebrafish-dev.cells.ucsc.edu> was used to assess expression of *sorcs2* and *grna*. Datasets labeled “a” and “b” represent developmental stages analyzed as experimental duplicates. The multi-cell transcriptome data from the laser-captured motor neurons of the spinal cord, the facial nucleus, and the cranial nuclei 3 and 4⁷² was downloaded from NCBI Gene Expression Omnibus (GEO, [https://www.ncbi.nlm.nih.gov/geo/accession number](https://www.ncbi.nlm.nih.gov/geo/accession%20number)) (GSE115706), and normalized expression values (RPKM) were extracted and plotted for *Sorcs2* and *Gm*. Deposited and normalized single-cell RNA sequencing data (<https://github.com/juliendelille/MouseSpinalCordAtlas>) from murine embryonic spinal cord⁶⁷ was downloaded from ArrayExpress (experiments/E-MTAB-7320/), and analyzed for the number of cells with detectable *Sorcs2* and *Gm* expression (Table S3; Figure S7).

Graphical visualization and statistics

All bright-field, chemiluminescent, fluorescent, and confocal images were mildly post-processed using either ImageJ¹⁰⁶ or Adobe Photoshop CC softwares to be suitable for data presentation and printed publication. All figures and drawings were arranged in Adobe Illustrator 2023. Drawings from Figures 4A and Figure S13 and the graphical abstract were prepared using BioRender.com. Videos from light-sheet microscope were prepared in Imaris v10.0.0. (Oxford Instruments). HCR3.0 and light-sheet reconstructions were exported as high-resolution snap-shots from the Imaris v.10_3D-viewer (Oxford Instruments).

All experiments were performed in biological and technical replicates as described above. All graphs were prepared in licensed GraphPad Prism software (versions 7/9/10), together with all statistical testing. We combined our academic knowledge in biostatistics with the GraphPad Prism algorithms allowing precise statistical evaluation. GraphPad Prism integrates complex, up-to-date statistical analyses together with detailed description and recommendation for data type, hypotheses, and checklist, including pre-selection of suitable statistical testing step-by-step. All data were first examined for statistical outliers and normal distribution. A p value of <0.05 was considered statistically significant. Corresponding statistical analysis used in each graph is stated in the table below. Data are presented as mean $\pm$ standard error of mean (SEM) except of Figure S12B which presents the data as median with 95% lower and upper confident intervals that is more suitable for the data type.

Statistical testing used in this study

Corresponding figure number	Statistical test
Figures 1I and 1J, Figures 2C–2E, 2G–2L, Figures 7E–7G	Kruskal-Wallis with Dunn’s test
Figure 1H	One-way ANOVA with Tukey’s test
Figure 2O	Mann-Whitney test
Figures 3F, 3H, 3I–3M, Figures S4A–S4F, S4I–S4M, S4P–S4W	Multiple unpaired t test with single pooled standard deviation for all rows
Figures 3G and Figures 4C–4H, Figures 6L and Figures 7I, 7N and 7O, Figure S2C, Figure S4G, Figure S4N, Figure S11M, Figure S12B	Unpaired two-tailed t test
Figures 4J–4M, Figures 5G–5I, Figures 7K and 7L, Figure S12A	Multiple unpaired t test
Figures 5K–5P	Two-way ANOVA with Dunnet’s test
Figure 6K	Two-way ANOVA with Sidák’s test
Figure 7M	One-way ANOVA with Fisher’s LSD test
Figures S2F–S2I	Two-way ANOVA with Tukey’s test